

THE SPECTRAL SHORTCUT: HOW A FAST CONTEXTUAL PATHWAY SUPPRESSES SPECTRAL LEARNING IN SERIAL SPECTRAL–SPATIAL MODELS

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ABSTRACT

Serial spectral–spatial models compress each pixel’s spectrum with a small encoder and classify with a wide spatial network that reads the encoded neighbourhood. When the neighbourhood predicts the label, does joint training starve the encoder? In a linear-encoder cross-entropy model with M replicated contextual readouts we prove that, at a matched fitting margin, the spectral coefficient moves by at most a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs, that the fitted model then fails under contextual reversal when its initial spectral coefficient is below half the margin, and that a $M^{-1/2}$ -normalized readout removes the width dependence without removing the failure: the contextual pathway’s training speed relative to the encoder is the competition parameter. Interventions on that speed in a ReLU convolutional model with two equally informative cues, and with measured tissue spectra in constructed neighbourhoods, show that an informative context suppresses the encoder’s spectral learning at matched fit at every width, that slowing the head restores it monotonically, and that the fitted head nonetheless predicts from context whenever the context is informative. The speed of the contextual pathway thus decides not whether the model fails under a context shift but whether the spectral cue remains recoverable from the encoder afterwards. On a production model, scalar curvature does not identify the mechanism and an earlier apparent freezing advantage was protocol-confounded; we recommend no mitigation.

1 INTRODUCTION

Infrared and quantum-cascade-laser histology classify tissue pixel by pixel from a spectrum of several hundred channels, and the models that do so are serial: a small encoder compresses each pixel’s spectrum, and a much larger spatial network reads the encoded neighbourhood (Berisha et al., 2019; O’Leary et al., 2026). The same shape appears wherever a per-token encoder feeds a contextual model. Practitioners often pretrain or freeze the spectral stage, and a companion tokenization study (anonymous, under review) reports a frozen random encoder competing with a jointly trained one. Whether joint training genuinely starves the encoder, and when that matters, is not settled by such comparisons: the spatial stage can predict the label from the neighbourhood, and a model that learns to do so may never need the spectrum.

This paper makes one claim and tests it at three levels. *In a serial spectral–spatial model, increasing the effective training speed of the contextual pathway suppresses learning of an equally predictive spectral cue; at matched training fit this produces greater reliance on context and failure when that context changes.* We prove it in a linear-encoder model with a replicated contextual readout (Section 3); we predict that it transfers to any serial model whose contextual pathway learns fast relative to the encoder, with head width in the standard parameterization and the head’s learning rate as two speed controls and readout normalization as the control that removes the width dependence; and we test that prediction with interventions in a nonlinear convolutional model on synthetic data (Section 4) and with measured tissue spectra in constructed neighbourhoods (Section 5), in each case with a control that removes the proposed cause.

Contributions.

- **Theorem 1.** For the serial cross-entropy model with M replicated contextual readouts, gradient flow from general initialization moves the spectral coefficient at the matched fitting margin by at most a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs; the fitted model misclassifies every example under contextual reversal when the initial spectral coefficient is below half the margin, with expected error tending to $\Phi(m/2\sigma)$ over isotropic Gaussian initialization; and a $M^{-1/2}$ -normalized readout removes the width dependence, though not the failure.
- **Synthetic intervention.** In a ReLU convolutional head over a linear encoder, with two cues of equal oracle information but unequal initial accessibility, an informative context suppresses the encoder’s spectral learning at matched fit at every width; slowing the whole head restores it monotonically in every seed, a normalized readout reduces it at large width, and an uninformative context removes it. The fitted head’s reliance on context does not respond to any of these controls: the contextual pathway’s speed sets how much spectral structure the encoder retains, hence what readout retraining can recover, not whether the shortcut is used.
- **Real spectra.** With measured centre spectra and their recorded labels in constructed neighbourhoods, the same reliance appears in every arm; encoder suppression is complete in the regime where the encoder must learn the contrast and, at the cue amplitude that makes the context accessible, insensitive to the head’s rate, while in the regime where a random encoder already reads the contrast there is nothing to suppress. A readiness scan of real class pairs is itself a result: which regime a pipeline is in depends on preprocessing and bottleneck.
- **Scope.** We report why scalar curvature on a production model does not identify the mechanism and why an earlier apparent freezing advantage was protocol-confounded (Section 6, Appendix E); no mitigation is recommended.

Related work. That one predictive feature can suppress the learning of another is gradient starvation (Pezeshki et al., 2021), and a preference for simpler features that fails under shift is simplicity bias (Shah et al., 2020); modality competition in multimodal training (Wang et al., 2020; Peng et al., 2022; Huang et al., 2022; Du et al., 2023) studies the same imbalance between input streams. Our contribution is narrower and, we think, more specific: an explicit finite-fit account of selective *encoder* learning in a *serial* system, with a reversal failure and a normalization control, tested by interventions on training speed rather than inferred from accuracies. The mathematics uses the balance invariant of gradient flow in layered linear models (Du et al., 2018; Saxe et al., 2014; Yun et al., 2021) and sits next to lazy-training analyses in which a wide component barely moves (Chizat et al., 2019; Yang & Hu, 2021); the width-dependent gain of a replicated readout is the mechanism, and the normalized readout is the corresponding parameterization control. Curvature and its growth with width and normalization (Karakida et al., 2019; van Laarhoven, 2017) are the reason we measure learning outcomes rather than curvature (Appendix E). Frozen or retrained feature extractors in practice (Kirichenko et al., 2023; Zhang et al., 2022; Coil & Cheney, 2025) are the setting in which the question arises; we do not evaluate them as remedies.

2 MODEL, TASK, AND WHAT COUNTS AS FAILURE

Serial spectral–spatial model. An input is an image whose pixel p carries a spectrum $x_p \in \mathbb{R}^S$ and a label $y_p \in \{-1, +1\}$. A spectral encoder $f_\theta : \mathbb{R}^S \rightarrow \mathbb{R}^K$ acts on each pixel separately, and a spatial model g_ϕ of width M reads the encoded neighbourhood of p to produce a logit F_p . The encoder is linear, $f_\theta(x) = Wx$ with $W \in \mathbb{R}^{K \times S}$, in the theorem and in the experiments; the spatial model is a replicated readout in the theorem and a ReLU convolutional head in the experiments. Training minimizes the mean margin loss $\mathcal{L} = \frac{1}{N} \sum_p \log(1 + e^{-y_p F_p})$ by gradient flow or plain gradient descent with one learning rate for every parameter, so that any difference in training speed between the two modules comes from the model, not from the optimizer.

Two cues. The training distribution carries the label in two places. The *spectral cue* lies in the pixel’s own spectrum along a direction the encoder must discover: a randomly initialized encoder attenuates it. The *contextual cue* lies in the neighbourhood along directions of large amplitude:

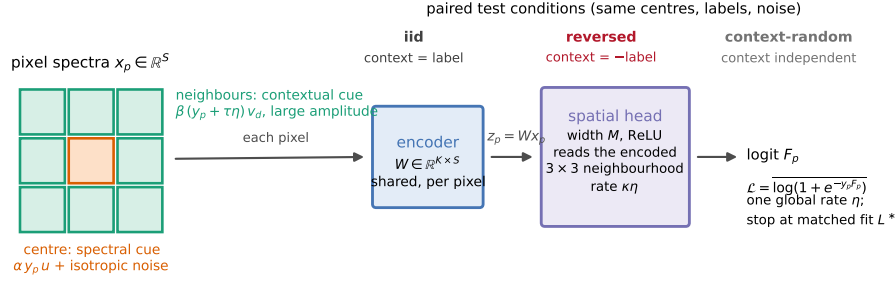


Figure 1: The serial model and the two cues. Each pixel’s spectrum passes through the shared encoder; the spatial model reads the encoded 3×3 neighbourhood. The spectral cue (own pixel, small amplitude along an unknown direction) requires encoder alignment; the contextual cue (neighbours, large amplitude along directions the random encoder already passes) does not. Evaluation under iid, reversed and context-random neighbourhoods measures reliance and failure.

a randomly initialized encoder passes it, so a trained readout can use it at once. Both cues are calibrated to carry the same information about the label (equal oracle accuracy), so the model’s preference cannot be attributed to one cue being more predictive. In the theorem this is the pair (a_0, v_0) : the initial spectral coefficient is small relative to the margin, the contextual coefficient is of order one. In the experiments it is a measured readiness asymmetry (Section 4).

Matched fit, spectral learning, reliance, failure. We compare models at their first attainment of a prespecified training-fit threshold: a common margin in the solvable model and mean training loss in the experiments. Spectral learning means adaptation of the designated spectral coefficient in the theorem and increased independently measured linear accessibility of the centre’s spectral cue in the experiments. We measure contextual reliance by changing the assigned neighbourhood signal while preserving the centre and its label, and recording the resulting change in predictions or risk. We call that reliance a failure when the shifted risk exceeds that of a separately trained spectral-only comparator at the same fitting threshold. The theorem supplies an exact reversal failure; the noisy experiments measure the corresponding risk difference without assuming a zero-error reference. This definition separates a model’s use of context from harm caused by that use, and separates information available in an encoder from information used by its trained head.

Three test conditions. Every evaluation set is generated three ways from the same centre pixels, labels and noise: *iid* (context agrees with the label as in training), *reversed* (the neighbourhood carries the opposite label), and *context-random* (the neighbourhood carries an independent label, so context is present but uninformative). Accuracy under reversal measures reliance; accuracy under context-random measures what the model can do from the spectrum alone in the presence of context.

3 MAIN RESULT: SELECTIVE SPECTRAL LEARNING AT MATCHED FIT

We state the mechanism in the simplest serial model that has both cues, a shared encoder, and a replicated contextual readout. Training inputs are two sites: the pixel’s own spectral coordinate S and a contextual coordinate C , both equal to the label. The encoder $W = (a, v)$ reads each site with one coefficient; the head adds the spectral coordinate directly and the contextual one through M replicated readouts. Replication is the only way width enters. The statement combines the serial instance, its general-initialization corollary and the readout-normalization proposition of Appendix A.

Theorem 1 (Selective spectral learning at matched fit). *Let Y be uniform on $\{-1, 1\}$, $x_1 = (S, 0)$, $x_2 = (0, C)$, and $S = C = Y$ in training. The shared encoder $W = (a, v)$ produces $z_1 = aS$, $z_2 = vC$; the head is $F = z_1 + bz_2$, $b = \sum_{j=1}^M \beta_j$. For integer $M \geq 1$, train (W, β) by unit Euclidean gradient flow on $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF})$, without other optimizer terms, from fixed $a_0 \in \mathbb{R}$, $v_0 \neq 0$, $\sum_j \beta_{j0} = 0$. Fix $m > 0$ and $T_m = \inf\{t \geq 0 : q_J(t) \geq m\}$, $q_J = a + bv$; thus $T_m = 0$ if $a_0 \geq m$.*

(a) **Suppression.** If $a_0 < m$, set $\Delta = m - a_0$, $u_M = a(T_m) - a_0$. Then $T_m < \infty$ and

$$0 < \frac{u_M}{\Delta} \leq \frac{1}{1 + Mv_0^2}, \quad Mu_M \rightarrow \frac{\Delta}{v_0^2} \quad (M \rightarrow \infty).$$

The spectral-only head $F = aS$ requires update Δ . If q_F instead trains only β from the same initialization with W frozen, then $q_J \geq q_F$ on $[0, T_m]$ and $\sup_{0 \leq t \leq T_m} (q_J - q_F) \leq C_0/(p_0 M v_0^2)$, where $C_0 = 1 + 2\Delta^2/v_0^2$ and $p_0 = (1 + e^{-a_0})^{-1}$.

(b) **Reversal.** On $S = Y, C = -Y$, the stopped margin is $2a(T_m) - m$ when $a_0 < m$, and a_0 otherwise. Error is one if $a_0 < m/2$ and $Mv_0^2 > m/(m - 2a_0)$; it is zero for $a_0 \geq m/2$. The spectral-only comparator has zero error for every a_0 . For $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, with zero-sum readouts and the same stopping rule, $\lim_{M \rightarrow \infty} \mathbb{E}_{W_0} \text{Err}_{\text{rev}} = \Phi(m/(2\sigma))$, where Φ is the standard normal CDF.

(c) **Normalization.** Replace b by $M^{-1/2} \sum_j \gamma_j$, with $\sum_j \gamma_{j0} = 0$ and the same (a_0, v_0) and unit rates. The aggregate (a, v, b) dynamics equal those at $M = 1$ for every width.

Remark 1 (Special initialization). For $(a_0, v_0) = (0, 1)$, the scalar-logit GGN blocks $G_{\theta\theta}, G_{\phi\phi}$, $\theta = (a, v)$, $\phi = \beta$, give $\mathcal{D}_{\text{curv}}(0) = M$ (generally Mv_0^2), and $\Psi(m)/(M + 1 + 2m^2) \leq T_m \leq \Psi(m)/(M + 1)$, where $\Psi(m) = m + e^m - 1$.

Proof sketch. Dividing the flow by $\dot{a} = \varrho > 0$, $\varrho = (1 + e^{q_J})^{-1}$, gives $dv/da = b$ and $db/da = Mv$, hence with $u = a - a_0$

$$v = v_0 \cosh(\sqrt{M}u), \quad b = \sqrt{M}v_0 \sinh(\sqrt{M}u), \quad q_J = a_0 + u + \frac{\sqrt{M}v_0^2}{2} \sinh(2\sqrt{M}u),$$

the balance invariant of gradient flow (Du et al., 2018). The fitting equation $q_J(u_M) = m$ is strictly increasing in u and $\sinh x \geq x$ gives the update bound and its limit; a transformed-margin comparison with the frozen path gives the finite-horizon gap. The sign of the reversed margin $2a(T_m) - m$ and dominated convergence give (b). For (c), differentiating the aggregate readout gives $\dot{b} = v\varrho$, independent of M . Full proofs are in Appendix A.

What the theorem says, and what it does not. Replication acts as an effective learning rate on the contextual readout. At the matched margin the spectral coefficient has moved by at most a fraction $1/(1 + Mv_0^2)$ of the update the spectral-only model needs; the coefficient itself can be informative at initialization and remain so. It is the *adaptation* that is suppressed. When the initial spectral coefficient is below half the margin, the fitted joint model relies on context strongly enough that contextual reversal misclassifies every example, while a spectral-only comparator trained to its own hitting time of the same margin is correct. Two comparators appear: the spectral-only model is trained to its own first hitting time from the same spectral initialization, whereas the frozen comparator is evaluated at the joint path’s elapsed times through T_m . Part (c) removes the width dependence only: the normalized system is the $M = 1$ system, which for $(a_0, v_0) = (0, 1)$ still has reversal error one for every $m > 0$. Whether normalization also reduces reliance in a trained network is an empirical question, asked in Section 4, not a consequence of (c). Finally, the same calculation with each contextual readout trained at rate κ instead of unit rate gives $\dot{b} = \kappa M v \varrho$ and the bound $1/(1 + \kappa M v_0^2)$: the relative training speed of the contextual pathway, not width as such, is the competition parameter. This is the quantity the experiments manipulate.

Numerical check. The closed forms were compared with a full $(2 + M)$ -parameter integration over 525 non-trivial initialization–width–margin cases ($M \in \{1, \dots, 4096\}$, $m \in \{2, 2.9, 4\}$): the maximum relative discrepancy over the four endpoint quantities $u_M, v(T_m), b(T_m), T_m$ is 1.3×10^{-11} , and no bound in (a)–(b) is violated in any case; the Gaussian sweep gives expected reversal error 0.8207, 0.8380, 0.8397 at $M = 64, 1024, 16384$ against the limit $\Phi(1) = 0.8413$ for $m = 2$, $\sigma = 1$ (Appendix B). This checks the algebra; it is not evidence for the claim beyond the model.

4 EXACT-MODEL CHECK AND SYNTHETIC INTERVENTION

Experiment 1: the exact model. The closed forms of Theorem 1 were checked against a full $(2 + M)$ -parameter integration of the flow over 525 initialization–width–margin cases and a Gaussian

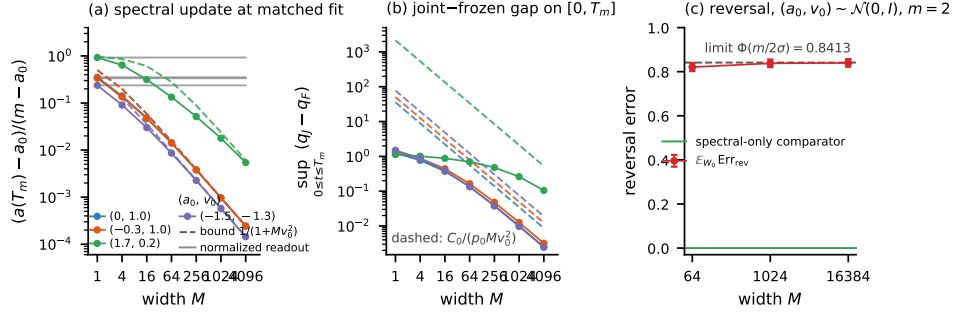


Figure 2: The exact mechanism (Theorem 1; $m = 2.9$ unless stated). (a) Spectral update fraction u_M/Δ at the matched margin against width M for four initializations (a_0, v_0) (solid) with the bound $1/(1 + Mv_0^2)$ (dashed) and the normalized readout of part (c) (grey, flat at the $M = 1$ value). (b) Joint-minus-frozen margin gap $\sup_{0 \leq t \leq T_m} (q_J - q_F)$ with the bound $C_0/(p_0 M v_0^2)$ (dashed). (c) Expected reversal error over $(a_0, v_0) \sim \mathcal{N}(0, I_2)$ against M for $m = 2$, with the limit $\Phi(m/2)$; the spectral-only comparator has error zero.

initialization sweep (Section 3, Appendix B); every bound holds in every case. This is a check of the algebra. The question the rest of the paper asks is whether the mechanism survives when the cues are noisy, the routing is learned and the head is nonlinear.

Experiment 2: design. Pixels carry i.i.d. labels $y_p \in \{-1, +1\}$ on 16×16 images with $S = 256$ spectral channels, so the spectral cue cannot be denoised by averaging neighbours. The spectral cue is $\alpha y_p u$ in the pixel’s own spectrum, buried in isotropic noise of unit variance, with u a fixed unit direction the encoder must discover. The contextual cue places y_p in each of the eight neighbours of p , along eight fixed orthogonal directions $v_1, \dots, v_8 \perp u$ of amplitude $\beta = 5$ with context noise $\tau \eta$: the information about y_p sits only in the neighbourhood, because p ’s own v -coordinates carry its neighbours’ labels. Equal information: $\alpha = 1.645$ gives an oracle spectral accuracy of 0.95, and $\tau = 1.70$ is calibrated so that the oracle contextual accuracy (a linear discriminant on the 3×3 window of the eight projections) is 0.950; both values are recorded before any training run (Appendix C). Readiness: through a random encoder $W \in \mathbb{R}^{12 \times 256}$ the same discriminant reads the contextual cue at 0.905–0.914 and the spectral cue at 0.608–0.653 (three seeds), because the random projection preserves the signal-to-noise ratio of a cue whose noise shares its direction and attenuates by about $\sqrt{K/S}$ one whose noise is isotropic. This is the nonlinear analogue of (a_0 small, $v_0 = O(1)$): the contextual cue is initially accessible to a trained readout; the spectral cue is not. The model is the linear encoder followed by a ReLU convolutional head (3×3 convolution to width M , ReLU, 3×3 convolution to one logit, circular padding), trained by full-batch gradient descent on 256 images (65,536 pixels) with one learning rate $\eta = 10^{-3}$ for every parameter, chosen as the largest rate with a monotone loss at the widest head. Runs are compared at the first step with training loss below $L^* = 0.30$, with further snapshots at 0.6, 0.5, 0.4, 0.2, 0.15, 0.10.

Arms and measures. Standard parameterization at widths $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; a normalized readout ($M^{-1/2}\gamma$ with $\gamma = O(1)$ and the same function at initialization, the control of Theorem 1(c), which under plain gradient descent equals a readout trained at rate $1/M$); training with an uninformative context (the same cue carrying an independent label: the proposed cause removed) at $M \in \{2, 8, 128, 512\}$; a readout learning-rate multiplier and, following the design review, a whole-head multiplier $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$, which in the theorem gives the bound $1/(1 + \kappa M v_0^2)$; and a frozen encoder. Three seeds each. At every snapshot we record the encoder’s linear accessibility of the spectral cue (a discriminant fitted on independent spectral-only data and evaluated on a disjoint set, reported as the gain over its initial value 0.647), the exact population counterpart $h_u(W) = \|P_{\text{row}(W)} u\|^2$ (the optimal probe has accuracy $\Phi(\alpha \sqrt{h_u})$; $h_u = 0.055$ at initialization), the encoder’s displacement, and accuracies on paired test sets that share labels and noise and differ only in the assigned context: iid, reversed, context-random, spectral-only, context-only.

Results: suppression. At L^* every jointly trained model with an informative context leaves the encoder’s spectral accessibility near its initial value: the probe gain is 0.04 at $M = 2$ and 0.002 at $M = 2048$ (means over three seeds; h_u 0.09 $\rightarrow$ 0.05 against 0.055 at initialization), accuracy under reversal is 0.12–0.13, and accuracy with an uninformative context is at chance (0.50). The same model trained with an uninformative context aligns its encoder before reaching the same loss (probe gain 0.25–0.30; h_u 0.56–0.86) and survives reversal (0.88–0.90). A frozen encoder gives reversal accuracy 0.12–0.14 (Table 4).

Results: the registered predictions. We report P1–P5 as registered (Appendix C). P1 fails: the width trend in encoder alignment and reversal accuracy is monotone in two seeds but not the third, and its magnitude is negligible, because L^* lies in a regime where the context fits before any spectral adaptation at every width, including $M = 2$. P2 fails as stated (the normalized readout also shows a small trend in two seeds), although its second clause holds: at $M = 2048$ the normalized readout leaves more encoder energy on the spectral direction than the standard one in every seed. P3 fails its literal cutoff on the encoder’s energy fraction while its substance holds: with the cause removed the encoder exposes the cue at every width (probe 0.89–0.94), the energy fraction merely spreading over more head directions as M grows. P4 holds for spectral learning (alignment and probe decrease with the readout multiplier in every seed) and fails for reversal accuracy, which rises by 0.02. P5 holds. A flat curve in a saturated regime does not falsify competition; it shows that the registered threshold could not resolve it.

Results: where the width and speed dependence appears. At the labelled secondary threshold 0.15, just above the population optimum of the contextual cue alone, spectral learning decreases with width in every seed (probe gain 0.22 at $M = 2$, 0.12 at $M = 8$, 0.06 at $M = 128$, 0.03 at $M = 2048$; Spearman -0.86 , -0.96 , -0.86), and the normalized readout is less suppressed at large width (0.10 at $M = 2048$ against 0.03; Table 5). The whole-head multiplier moves the same quantity at L^* itself: probe gain 0.02, 0.09, 0.21 and h_u 0.07, 0.14, 0.41 for $\kappa = 1, 1/16, 1/256$ (means of three seeds, monotone in every seed; Table 7), in the direction of the theorem’s bound $1/(1 + \kappa M v_0^2)$; at $\kappa \leq 1/4096$ the head is too slow to reach L^* within the step budget, and the encoder ends with $h_u = 0.70$. The readout-only multiplier acts in the same direction but weakly (Table 8).

Results: reliance does not follow speed. Accuracy under reversal at L^* stays between 0.11 and 0.21 across every width, every κ including the extension, the normalized readout and the readout multiplier, and accuracy with an uninformative context stays at chance; only training without an informative context removes the reliance. The encoder can be made to expose the cue almost fully (h_u 0.70, probe 0.91) while the head still predicts from context. In the theorem’s terms the fitted models sit in the regime $2a(T_m) < m$: the contextual pathway’s speed relative to the encoder controls how much of the spectral cue the encoder learns before the fit is reached; the head’s gain for the large-amplitude contextual cue, which slowing the whole head leaves unchanged, controls which cue the fitted head uses. The probe is a linear readout on the frozen encoder, so it measures what readout retraining without the shortcut can recover (Kirichenko et al., 2023): a fast wide head leaves 0.65–0.75 recoverable spectral accuracy, a slow head, a small width, a normalized readout at large width or an uninformative context leave 0.87–0.94.

Amendments. Widths 2 and 4, the fractional readout multipliers and the whole-head arm were added after a one-seed pilot and the design review, before the corresponding runs; the κ bracket was extended downward for the reversal outcome after the seed-0 pilot (Appendix C). The original five-width predictions are reported above as registered.

5 REAL CENTRE SPECTRA WITH CONSTRUCTED CONTEXT

What this experiment is. We test contextual competition using measured centre spectra and their recorded labels, with a synthetic class-associated spatial intensity pattern along training-estimated principal directions in the neighbourhood. It is a real-centre-spectrum experiment with constructed context, not a validation on natural tissue context: it asks whether the mechanism operates when the spectral cue is a real, high-dimensional class contrast, and it cannot show that natural neighbourhoods produce it.

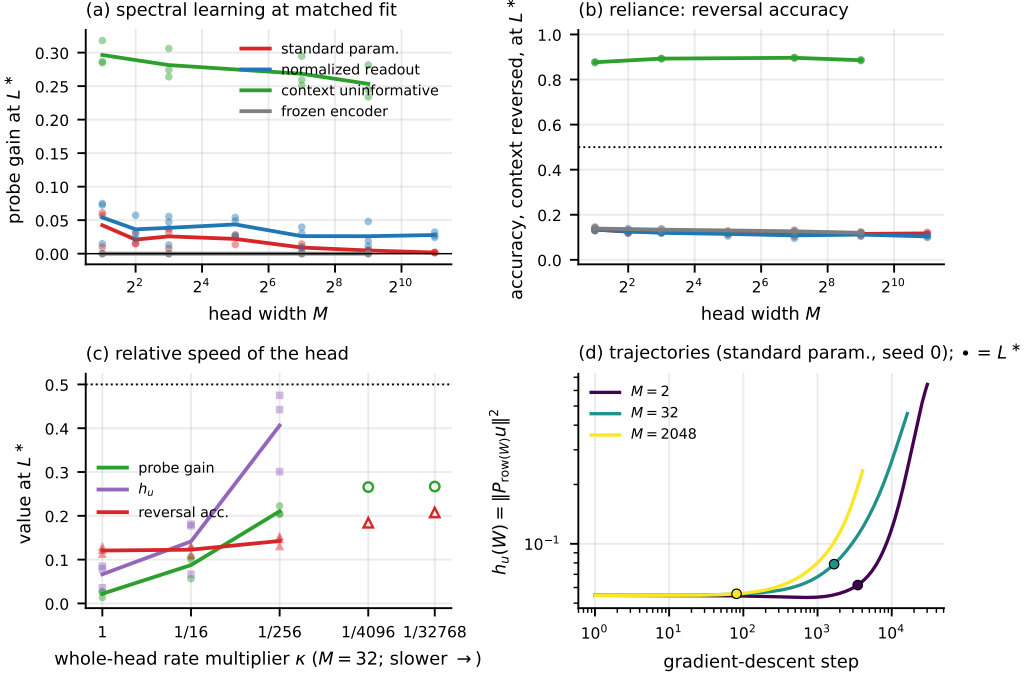


Figure 3: Experiment 2 at matched fit $L^* = 0.30$ (three seeds; points are seeds, lines are means). (a) Spectral probe gain over initialization against head width for the standard parameterization, the normalized readout, the uninformative-context control and the frozen encoder. (b) Accuracy under context reversal against width. (c) Whole-head rate multiplier κ at $M = 32$: probe gain and reversal accuracy. (d) Trajectories of h_u for three widths with the fit thresholds marked.

Data and readiness. Breast-tissue QCL micro-array cores, fold 0 of a five-fold patient-level split; the binary task Cancer Epithelium versus Cancer-Associated Stroma on 942-channel pixel vectors; 24,000 class-balanced training centres, evaluation on held-out validation patients with test patients reported separately (Appendix D). A readiness scan (Table 9) shows that with per-feature standardization a random encoder $W \in \mathbb{R}^{12 \times 942}$ already reads this contrast at 0.955 against an oracle of 0.969, because the class-mean contrast lies along the top principal directions; PCA whitening estimated on the training patients reduces the random-encoder probe to 0.54 while the oracle stays at 0.944. We therefore run two regimes on the same pixels and labels: *ready* (standardized inputs, nothing for the encoder to learn) and *unready* (whitened inputs, the theorem’s case).

Construction. Each patch is a real centre with eight real donor spectra drawn independently of the centre label from the same split side; donor d carries $\gamma_d(c s + \tau \eta_d)$ along the d th principal coordinate, where c is the centre’s label, $s = \pm 1$ or c is replaced by an independent label, and γ_d is large relative to the donors’ natural variation ($3\sqrt{\lambda_d}$ standardized; 30 in whitened units, with a $\gamma = 10$ variant added after the first results). τ is calibrated per regime so that the neighbourhood oracle equals the centre-only oracle (0.969 ready, 0.948 unready); through a random encoder the neighbourhood reads at 0.94–0.99 in both regimes. The model is the linear encoder over the nine spectra and a ReLU head of width M over the encoded patch; training and thresholds are those of Section 4; the probe is fitted on disjoint training pixels and evaluated on validation centres.

Results. In the *ready* regime the probe is 0.91–0.94 before training and does not move at L^* in any arm (gain -0.015 to $+0.035$), yet every jointly trained model predicts from context only: reversal accuracy 0.08–0.10 and context-random accuracy at chance for widths 8–512 and for $\kappa = 1, 1/16, 1/256$ alike, all fitting within 70–230 steps; the uninformative-context comparator reaches reversal 0.93 and context-random 0.92, and the frozen encoder 0.09 (Table 13). Nothing can be starved here; the reliance is head-level competition between a large-amplitude cue and an $O(1)$ one, which the theorem does not address, and it is complete.

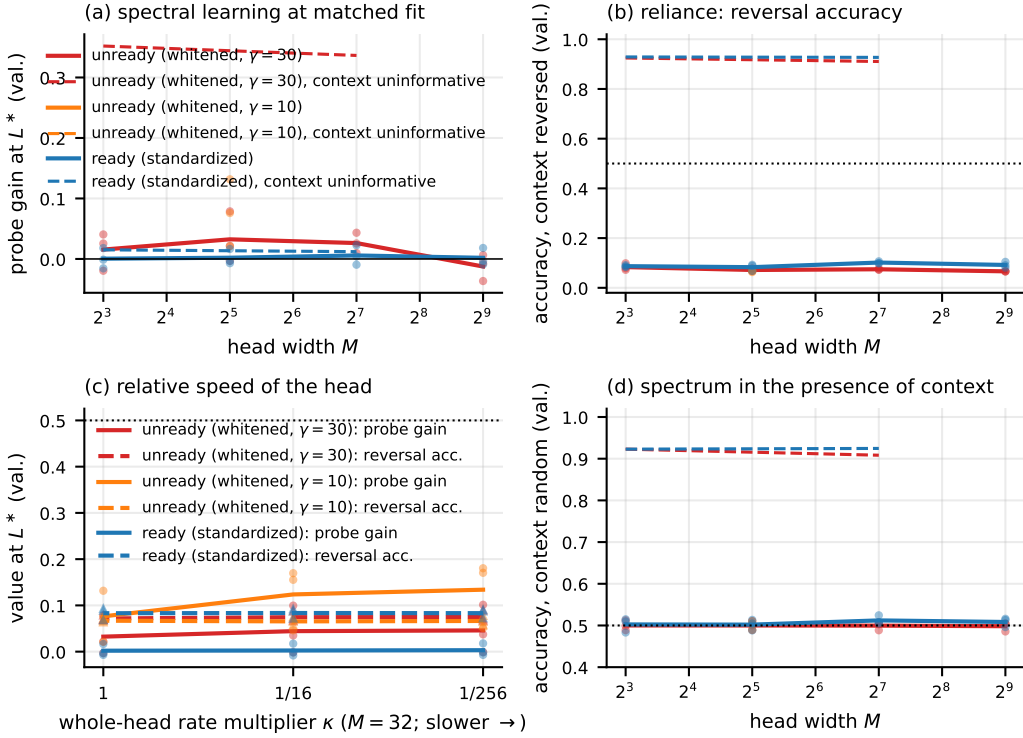


Figure 4: Experiment 3 at $L^* = 0.30$ on validation patients (three seeds; points are seeds). (a) Spectral probe gain against width in the two regimes, with the uninformative-context comparator dashed. (b) Accuracy under context reversal. (c) Whole-head rate multiplier at $M = 32$. (d) Accuracy with an uninformative context: what the model does from the spectrum in the presence of context.

In the *unready* regime the probe starts at 0.56–0.60 and at L^* has gained at most 0.05 in any arm, with reversal 0.07 and context-random at chance; the fit takes 69–185 steps even at $\kappa = 1/256$ (Table 11). Slowing the head does not help the encoder here because, at amplitude 30, the contextual cue dominates the encoder’s own gradient: the encoder learns the context directions before the whitened, high-dimensional class contrast, in the theorem’s terms a large v_0 relative to a_0 at the encoder rather than a fast readout. The comparators show what the encoder can do with the same real spectra when the cause is removed: trained with an uninformative context it reaches L^* only after 15,000–17,000 steps but with probe gain +0.35 (probe 0.92–0.93), reversal accuracy 0.91–0.93 and context-random accuracy 0.91–0.92; the frozen encoder gives 0.08. Excess shifted risk of the joint models over that comparator is 0.85. A variant with the cue amplitude reduced to $\gamma = 10$, the scale of Section 4, was added to test whether the head’s rate recovers its effect once the cue no longer dominates the encoder’s gradient; the prediction was recorded before the runs. It does: the fit now takes 530–1,800 steps, and the probe gain at L^* rises with a slower head in every seed (0.02, 0.05, 0.05; 0.13, 0.17, 0.18; 0.08, 0.16, 0.17 for $\kappa = 1, 1/16, 1/256$ in seeds 0–2), while reversal accuracy stays at 0.06–0.08 and the uninformative-context comparator again reaches a gain of 0.35 and reversal 0.92 (Table 12).

Reading. With measured spectra the outcome matches the synthetic experiment on reliance and sharpens the account of suppression: the fitted head uses the constructed context whenever it is informative, at every width and rate; whether the encoder learns the spectral contrast before the fit depends on the balance of gains at the encoder, set jointly by the head’s relative speed (Section 4) and by the contextual cue’s amplitude relative to the spectral one. Both are the theorem’s $\kappa M v_0^2$ seen from different sides. Excess shifted risk relative to the uninformative-context comparator is 0.85 in both regimes.

6 LIMITATIONS AND CONCLUSION

Conclusion. We prove, in a serial linear-encoder model, that a contextual pathway trained faster than the spectral encoder suppresses the encoder’s update at matched fit, that the fitted model then fails under contextual reversal while a spectral-only comparator does not, and that readout normalization removes the width dependence of this effect without removing the failure. In a ReLU convolutional head over a linear encoder on synthetic data, and with measured tissue spectra in constructed neighbourhoods, we find the corresponding suppression at every width and the corresponding failure under context shift whenever the context is informative, with excess shifted risk of about 0.75 (synthetic) and 0.85 (real spectra) over the uninformative-context comparator. What transfers with a qualification is the role of speed: the head’s relative rate controls how much of the spectral cue the encoder learns before the fit, not whether the fitted head uses the shortcut, and where the contextual cue dominates the encoder’s own gradient even that control disappears.

Limitations. The theorem has a linear encoder, noiseless cues and a fixed spectral skip coefficient; the experiments add noisy cues and jointly learned routing, but keep a linear encoder and one head family each. The contextual cues are constructed, in the synthetic data by design and in the real-spectrum experiment explicitly, so neither experiment shows that natural tissue neighbourhoods produce the effect. Matched mean training loss is an experimental analogue of the theorem’s matched margin, not the same stopping rule, and the fit threshold is a convention for comparison, not a training recipe. The initial accessibility asymmetry that the mechanism needs is set by the bottleneck and the input statistics: where a random encoder already exposes the spectral cue, as it does for one real class pair at bottleneck twelve (Section 5), the theorem’s $a_0 \geq m/2$ case applies and no failure is predicted. Plain gradient descent with one global rate is the only optimizer studied.

What curvature diagnostics do not tell us. On a production spectral–spatial segmentation model we measured the initialization curvature of the two parameter blocks and found the scalar ratio inverted (0.46 ± 0.21 at width 192) because the inputs are effectively rank one (effective rank 1.07 of 942), and found its growth with width to be a normalization-gain effect (first-head-convolution slope $+0.61$ with batch statistics against $+0.006$ with both head BatchNorm layers at fixed statistics), while the class-contrast direction carries $12\text{--}28\times$ less curvature than the dominant input direction (Appendix E). Scalar curvature therefore does not identify the mechanism, and directional curvature is not a measurement of impaired learning; this is why the present paper measures learning outcomes under interventions rather than curvature.

What a matched retraining study did not show. At width 48 and three seeds, an earlier apparent advantage of freezing the spectral encoder in that production model came from a comparison that confounded freezing with gradient-clipping scope and normalization-parameter treatment. Once those differences were removed, its nominal final-window advantage was absent, checkpoint selection reversed the ordering within the original runs, and neither the within-protocol freezing contrasts nor the change in their gap establishes a population effect (paired 90% intervals in Appendix E). That study cannot identify an architectural shortcut and does not support a mitigation recommendation; we make none.

Open problem. Whether the mechanism operates in production pipelines trained from scratch with natural context statistics, where readiness, cue balance and optimizer all differ from the controlled settings here, is the question these results leave open.

AI USE STATEMENT

Generative AI tools (large language model assistants) were used throughout this work: for literature search; for drafting and editing text; for writing and checking code, including the numerical verification scripts of Appendix F; and for adversarial review of the proofs and of the claims made about the measurements. Every proof was independently re-derived by the authors, and every number reported in the paper was recomputed by the authors from the recorded artifacts. All AI-generated text and code were reviewed by the authors, who take full responsibility for the content of this paper, including any remaining errors.

REPRODUCIBILITY STATEMENT

Code, configurations and audit scripts, with seeds and exact commands, are provided in the anonymized repository accompanying this submission. Every number in the paper traces to a named results file; Appendix F lists the verification scripts and their outcomes, including the two that do not exit cleanly. Complete hypotheses and proofs are in Appendix A, with the numerical verification in Appendix B. Generators, calibration, architectures, data, patient splits and training protocols are described in Appendices C, D and E.

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A PROOFS OF THEOREM 1

Theorem 1 combines three statements proved below in their original form: the serial instance at the special initialization (Theorem A.1), its general and isotropic initialization (Corollary A.1), and the readout normalization (Proposition A.2). Part (a) of Theorem 1 is (10) together with the finite-horizon gap (12); part (b) is the reversal statement of Corollary A.1 including (13); part (c) is Proposition A.2 with $\alpha_M = M^{-1/2}$, whose proof applies to any zero-sum initialization of the normalized readouts because only the aggregate b enters the flow. The rate- κ variant quoted in Section 3 follows by the same differentiation with $\dot{\beta}_j = \kappa v \varrho$. The special-initialization remark is (4) and the GGN display of Theorem A.1. Notation: $\varrho = (1 + e^q)^{-1}$ is the logistic residual, $\Psi(q) = q + e^q - 1$, and $G_{\theta\theta}, G_{\phi\phi}$ are the Gauss–Newton blocks of the scalar-logit loss in the parameter blocks $\theta = (a, v)$ and $\phi = \beta$.

A.0.1 THE SERIAL INSTANCE, ITS GENERAL INITIALIZATION, AND THE NORMALIZED READOUT

Proposition A.1 (Information retained by a fitting serial encoder). *If the predictions depend on X only through $Z = f_\theta(X)$ and a classifier $q_\phi(Y | Z)$ has population CE at most $\eta < \infty$, then $H(Y | Z) \leq \eta$ and $I(Y; Z) \geq H(Y) - \eta$. The same statement holds for the empirical distribution with a fixed model.*

Proof. CE is conditional entropy plus the nonnegative expected conditional KL divergence. Suppression of a selected coordinate or failure under a specified shift is therefore different from absence of all training-label information in the serial representation. $\square$

Theorem A.1 (Serial CE instance; review Theorem 5.1). *Let Y be uniform on $\{-1, 1\}$ and let training inputs be $x_1 = (S, 0)$, $x_2 = (0, C_{\text{ctx}})$ with $S = C_{\text{ctx}} = Y$. The shared linear encoder is $W = (a, v)$, giving $z_1 = aS$, $z_2 = vC_{\text{ctx}}$. For integer $M \geq 1$ define*

$$F = z_1 + \sum_{j=1}^M \beta_j z_2 = aS + bvC_{\text{ctx}}, \quad b = \sum_j \beta_j, \quad \theta = (a, v), \quad \phi = \beta.$$

All coordinates follow unit Euclidean gradient flow on the averaged binary CE $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF}) = \log(1 + e^{-q})$, $q = a + bv$, from $a_0 = 0, v_0 = 1, \beta_0 = 0$, with no clipping, decay or momentum. Fix $0 < \delta < 1/2$, $m = \log((1 - \delta)/\delta)$, and let T_m be the first time q reaches m . Write $\varrho = (1 + e^q)^{-1}$ and $\Psi(q) = q + e^q - 1$. Then:

$$\dot{a} = \varrho, \quad \dot{v} = b\varrho, \quad \dot{b} = Mv\varrho, \quad (1)$$

$$v = \cosh(\sqrt{M}a), \quad b = \sqrt{M} \sinh(\sqrt{M}a), \quad (2)$$

$$q = a + \frac{\sqrt{M}}{2} \sinh(2\sqrt{M}a), \quad \dot{q} = (M + 1 + 2b^2)\varrho. \quad (3)$$

The solution is global and reaches m in finite time, with

$$\frac{\Psi(m)}{M + 1 + 2m^2} \leq T_m \leq \frac{\Psi(m)}{M + 1}. \quad (4)$$

At that time, writing $a_M = a(T_m)$,

$$0 < a_M \leq \frac{m}{M + 1}, \quad b(T_m)v(T_m) \geq \frac{Mm}{M + 1}, \quad Ma_M \rightarrow m, \quad (5)$$

$$0 \leq v(T_m) - 1 \leq \frac{m^2}{2M}, \quad \|W(T_m) - W_0\| \leq \sqrt{\frac{m^2}{(M + 1)^2} + \frac{m^4}{4M^2}}. \quad (6)$$

For fixed m , a_M decreases strictly with M ; removing the contextual path entirely requires $a = m$ at the same fitting margin. Moreover,

$$\varrho(t) \leq \frac{1/2}{1 + (M + 1)t/4}, \quad t \geq 0. \quad (7)$$

In the stated parameter blocks and loss convention,

$$\lambda_{\max}(G_{\theta\theta}) = \varrho(1 - \varrho)(1 + b^2), \quad \lambda_{\max}(G_{\phi\phi}) = M\varrho(1 - \varrho)v^2,$$

so at initialization they equal $1/4, M/4$ and the disparity is M . For the separately trained frozen encoder $W = (0, 1)$, $\beta_F(0) = 0$, its margin obeys $\Psi(q_F(t)) = Mt$ and

$$0 \leq q(t) - q_F(t) \leq \frac{(1 + 2m^2)t}{2 + Mt/2} \leq \frac{2(1 + 2m^2)}{M}, \quad 0 \leq t \leq T_m. \quad (8)$$

Under the specified reversal $S = Y, C_{\text{ctx}} = -Y$, the joint model at T_m has error one; the fitted spectral-only comparator has error zero.

Proof. The training margin is the same on each example, so direct differentiation gives (1). Divide by $\dot{a} = \varrho > 0$ to get $dv/da = b$, $db/da = Mv$, which proves (2). It implies $b^2 = M(v^2 - 1)$ and (3). Since $0 < \dot{a} \leq 1/2$, a stays finite on every finite time interval; the phase formula then gives global existence. The strictly increasing map $Q_M(a) = a + (\sqrt{M}/2) \sinh(2\sqrt{M}a)$ maps $[0, \infty)$ onto itself and has a finite inverse at m . On this compact interval, $\dot{a} \geq \delta > 0$, so the hitting time is finite.

$\sinh x \geq x$ gives $Q_M(a) \geq (M + 1)a$, strictly for $a > 0$. Before fitting, $v \geq 1, 0 \leq bv = q - a \leq m$, hence $b \leq m$. The invariant gives $v - 1 = b^2/[M(v + 1)] \leq m^2/(2M)$. For fixed positive a the contextual term increases strictly with M ; invert to get strict decrease of a_M . Its upper bound implies $\sqrt{M}a_M \rightarrow 0$, and the expansion $m = (M + 1)a_M + O(M^2a_M^3)$ gives $Ma_M \rightarrow m$.

$\dot{\varrho} = -(M + 1 + 2b^2)\varrho^2(1 - \varrho)$ and $\varrho \leq 1/2$ imply $(d/dt)(1/\varrho) \geq (M + 1)/2$. Integrate from $\varrho(0) = 1/2$ for the envelope. Since $\Psi'(q) = 1/\varrho$, before fitting $\frac{d}{dt}\Psi(q) = M + 1 + 2b^2 \in$

$[M + 1, M + 1 + 2m^2]$; integration proves (4). The margin gradients are $Y(1, b)$ and $Yv\mathbf{1}_M$; the scalar logistic Hessian is $\varrho(1 - \varrho)$, giving the GGN.

For the frozen system $\dot{q}_F = M/(1 + e^{q_F})$. Thus $0 \leq \Psi(q) - \Psi(q_F) = \int_0^t (1 + 2b^2)ds \leq (1 + 2m^2)t$. Convexity gives a lower bound $\Psi'(q_F)(q - q_F)$ for this difference. As $q_F \leq e^{q_F} - 1$, $Mt = \Psi(q_F) \leq 2(e^{q_F} - 1)$, so $\Psi'(q_F) \geq 2 + Mt/2$, proving (8). The shifted margin is $a - bv = 2a_M - m < 0$: for $M > 1$ use (5), and for $M = 1$ use the strict inequality $Q_1(a) > 2a$. The spectral-only shifted margin is m . $\square$

Remark A.1 (Embedding, head, and initialization conventions). For logits $(F, 0)$, loss, residual norm and GGN agree, but the unprojected logit-Jacobian norm is $\sqrt{1 + b^2}$; the symmetric embedding makes the norm-product bound exact. A freely trained dense two-row head is not the parameter block $\phi = \beta$: its initial head eigenvalue is $M/2$, not $M/4$, while the encoder eigenvalue remains $1/4$ at the same initial function. Labels need not be balanced for the training ODE; balance is used in the information/noisy-accuracy statement below. The initial encoder orientation in Theorem A.1 is explicitly asymmetric; the next corollary removes that choice for W , not the routing/readout asymmetry.

Proposition A.2 (Readout normalization and representation interpretation). *Replace the contextual readout by $\alpha_M \sum_j \gamma_j z_j$, $\gamma_j(0) = 0$, at unit rates. Its reduced equation for $b = \alpha_M \sum_j \gamma_j$ is $\dot{b} = M\alpha_M^2 v \varrho$. In particular $\alpha_M = M^{-1/2}$ gives exactly the $M = 1$ dynamics for every M . At every finite M in the original theorem, the noiseless spectral coordinate $a_M S$ still determines Y . If it is instead observed as $Z_S = a_M Y + \sigma \xi$, with $\xi \sim \mathcal{N}(0, 1)$ independent, $\sigma > 0$ fixed, then Bayes accuracy is $\Phi(a_M/\sigma)$ and $I(Y; Z_S) \leq a_M^2/(2\sigma^2)$.*

Proof. Differentiate each readout and sum. The Gaussian likelihood-ratio test is thresholding at zero, giving the accuracy. For $Q = \mathcal{N}(0, \sigma^2)$, $\mathbb{E}_Y \text{KL}(P_{Z_S|Y} \| Q) = a_M^2/(2\sigma^2) = I(Y; Z_S) + \text{KL}(P_{Z_S} \| Q)$. This noise is a specified observation perturbation, not training noise. $\square$

Corollary A.1 (General and isotropic encoder initialization). *In the serial model let $m > 0$, $a_0 < m$, $v_0 \neq 0$, and allow any readout initialization with $\sum_j \beta_{j0} = 0$. Set $u = a - a_0$, $\Delta = m - a_0 > 0$. Then*

$$v = v_0 \cosh(\sqrt{M}u), \quad b = \sqrt{M}v_0 \sinh(\sqrt{M}u), \quad (9)$$

$$q = a_0 + u + \frac{\sqrt{M}v_0^2}{2} \sinh(2\sqrt{M}u),$$

$$0 < u_M \leq \frac{\Delta}{1 + Mv_0^2}, \quad Mu_M \rightarrow \frac{\Delta}{v_0^2}, \quad \frac{a(T_m) - a_0}{m - a_0} \rightarrow 0. \quad (10)$$

The fitting root decreases strictly with M and

$$|v(T_m) - v_0| \leq \frac{\Delta^2}{2M|v_0|^3}, \quad \|W(T_m) - W_0\| \leq \sqrt{\frac{\Delta^2}{(1 + Mv_0^2)^2} + \frac{\Delta^4}{4M^2|v_0|^6}}. \quad (11)$$

With $\Psi_{a_0}(q) = q - a_0 + e^q - e^{a_0}$,

$$\frac{\Psi_{a_0}(m)}{1 + Mv_0^2 + 2\Delta^2/v_0^2} \leq T_m \leq \frac{\Psi_{a_0}(m)}{1 + Mv_0^2}.$$

For a frozen encoder at (a_0, v_0) and the same initial readouts, $\Psi_{a_0}(q_F) = Mv_0^2 t$. If $p_0 = e^{a_0}/(1 + e^{a_0})$ and $C_0 = 1 + 2\Delta^2/v_0^2$, then

$$0 \leq q_J - q_F \leq \frac{C_0 t}{1 + e^{a_0} + p_0 Mv_0^2 t} \leq \frac{C_0}{p_0 Mv_0^2}, \quad 0 \leq t \leq T_m. \quad (12)$$

Under contextual reversal the fitted margin is $2a(T_m) - m$. For $a_0 < m/2$, error is one for all sufficiently large M , with sufficient condition $Mv_0^2 > m/(m - 2a_0)$. For $m/2 \leq a_0 < m$, the reversed classifier is correct at every width. If $a_0 \geq m$, define fitting to stop immediately, in which case it is also correct.

If $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, then $v_0 \neq 0$ almost surely; use the preceding stopping convention. The expected reversal error satisfies

$$\lim_{M \rightarrow \infty} \mathbb{E}_{W_0} \text{Err}_{\text{reversal}} = \Phi\left(\frac{m}{2\sigma}\right). \quad (13)$$

The same-threshold spectral-only comparator has zero reversal error.

Proof. Dividing the ODE by $\dot{a} > 0$ gives (9); only the initial readout sum enters. Each $\beta_i - \beta_j$ is conserved, so equal initial readouts are not needed for closure. Positivity of the contextual margin $bv = (\sqrt{M}v_0^2/2) \sinh(2\sqrt{M}u)$ and the sinh inequality prove (10); Taylor expansion at $\sqrt{M}u_M \rightarrow 0$ gives the limit. The invariant is $b^2 = M(v^2 - v_0^2)$; v keeps its sign and $|v| \geq |v_0|$. Since $bv \leq \Delta$, $|b| \leq \Delta/|v_0|$ and

$$|v - v_0| = \frac{b^2}{M(|v| + |v_0|)} \leq \frac{\Delta^2}{2M|v_0|^3}.$$

This finite-M inequality proves (11). Now $\dot{q} = (1 + Mv_0^2 + 2b^2)\varrho$ and $b^2 \leq \Delta^2/v_0^2$ on the fitting interval; integrate the transformed margin for the time bounds. Global existence follows as before from $\dot{u} \leq 1$ and the phase formula; every finite m is hit.

The transformed joint/frozen difference lies between zero and $C_0 t$. Also $q_F - a_0 \leq (e^{q_F} - e^{a_0})/e^{a_0}$ gives $\Psi'_{a_0}(q_F) \geq 1 + e^{a_0} + p_0 M v_0^2 t$. Convexity proves (12). For $a_0 < m/2$, use $a(T_m) \leq a_0 + \Delta/(1 + Mv_0^2)$ to get the finite-M reversal condition. For $a_0 > m/2$ the classifier is correct at every fitted width, since $u_M > 0$ gives $2a(T_m) - m > 2a_0 - m > 0$; equality $a_0 = m/2$ also gives a positive margin. Immediate stopping covers $a_0 \geq m$. Almost surely the limiting error is $1_{\{a_0 < m/2\}}$, apart from the zero-probability boundary and $v_0 = 0$ events. Dominated convergence gives (13). $\square$

Remark A.2 (Confidence dependence and limiting quantity). For $a_0 \neq 0$, $a(T_m)/a_0 \rightarrow 1$ but $a(T_m)/m \rightarrow a_0/m$; the vanishing quantity is the *update* fraction in (10). For $0 < \rho < 1$, a Gaussian tail and its maximum density give, with probability at least $1 - \rho$,

$$|a_0| \leq \sigma \sqrt{2 \log(4/\rho)}, \quad |v_0| \geq \frac{\rho \sigma}{2} \sqrt{\pi/2}.$$

Substitute these bounds in the finite-M inequalities for uniform confidence bounds. The dependence is severe: for fixed $(\Delta, v_0) \in (0, \infty) \times (\mathbb{R} \setminus \{0\})$, expansion of the root yields

$$Mu_M = \frac{\Delta}{v_0^2} \left[1 - \frac{1}{Mv_0^2} - \frac{2\Delta^2}{3Mv_0^4} + O(M^{-2}) \right].$$

Indeed substitute $u = A/M + B/M^2$ in the sinh equation and match its constant and M^{-1} coefficients. The expansion's remainder is pointwise, not already uniform in ρ . Its small- $|v_0|$ crossover requires control of $\Delta^2/(Mv_0^4)$; at the confidence cutoff this has scale $\Delta^2 \rho^{-4} \sigma^{-4}/M$, a fourth power of $1/\rho$, not a second. A rigorous uniform remainder can instead be bounded from the sinh series on this controlled argument. Do not present this asymptotic crossover as a universal necessary width.

Provenance and interpretation. Architecture- and initialization-dependent implicit bias is established in linear-network analyses (Yun et al., 2021; Moroshko et al., 2020); trajectory-level incremental learning is studied by Berthier (2023). Exact deep-linear dynamics (Saxe et al., 2014) and conservation of layer-norm differences (Du et al., 2018) supply the underlying tools. CE feature competition is central to Pezeshki et al. (2021). The serial example instantiates a known optimization-metric mechanism with finite-margin formulas, an encoder-update comparison and a specified reversal distribution. Replication is equivalent to a rate M on b , and normalization removes it; the function class does not grow. These particular calculations are not a claim of a new general implicit-bias mechanism, and finite-time or serial composition alone is not a novelty claim.

B EXACT-MODEL VERIFICATION

The closed forms of Appendix A were checked against a full $(2 + M)$ -parameter integration of the gradient flow. The tables below are reproduced from the verification record of the earlier

draft (code/experiments/toy_serial_ce.py, toy_serial_ce_isotropic.py); the maximum relative discrepancy over the four endpoint quantities in the 525 non-trivial isotropic cases is 1.26×10^{-11} , the maximum trajectory-identity discrepancy is 1.5×10^{-10} , and no bound is violated in any case. The quantities and error conventions are those stated in each caption.

B.0.1 SOLVABLE SERIAL INSTANCE: NUMERICAL VERIFICATION

Zero-context instance. Table 1 verifies the closed forms and the bounds of the solvable serial cross-entropy instance at $a_0 = 0$, $v_0 = 1$ and margin $m = \log 19 = 2.944$, from results/toy_serial_ce.csv. The solver column is a stiff ODE integration of the full $(2 + M)$ -dimensional state; a separate explicit-Euler gradient-descent path, run at four step sizes, has observed convergence order 1.0000004 and agrees with an independent numpy analytic-gradient implementation to 0 relative error.

Table 1: Solvable serial instance ($a_0 = 0$, $v_0 = 1$, $m = \log 19$): closed form against ODE solver, from results/toy_serial_ce.csv. The last block is the two-timescale displacement bound against the realised encoder displacement. Every relative discrepancy between closed form and solver in this table is at most 1.2×10^{-13} .

Quantity	$M = 16$	$M = 64$	$M = 256$	$M = 1024$
$\lambda_\theta(0)$	0.25	0.25	0.25	0.25
$\lambda_\phi(0)$	4.0	16.0	64.0	256.0
$\mathcal{D}_{\text{curv}}(0)$	16.0	64.0	256.0	1024.0
$a(T_m)$	0.142325	0.042094	0.011216	0.002857
suppression m/a_M	20.688	69.949	262.53	1030.7
$M a_M$ (limit m)	2.2772	2.6940	2.8712	2.9253
T_m	0.90784	0.28680	0.078763	0.020252
T_m lower bound $\Psi(m)/(M+1+2m^2)$	0.60992	0.25437	0.076345	0.020094
T_m upper bound $\Psi(m)/(M+1)$	1.2320	0.32222	0.081496	0.020434
$\sup_t (q_J - q_F)$	0.38232	0.13813	0.039802	0.010376
uniform gap bound	11.297	2.9547	0.74729	0.18737
$(M+1) \times \text{gap}$	6.4994	8.9787	10.229	10.635
$\ W(T_m) - W(0)\ _2$	0.219021	0.071050	0.019658	0.005064
displacement bound	1.370601	0.358465	0.090662	0.022732
bound / displacement	6.258	5.045	4.612	4.489
normalized-readout control: $a(T_m)$	1.026947	1.026947	1.026947	1.026947
normalized-readout control: $\ W(T_m) - W(0)\ $	1.177117	1.177117	1.177117	1.177117
residual envelope excess on $[0, 20T_m]$	0	0	0	0
readout spread $\max_j \beta_j - \min_j \beta_j$ at T_m	0	0	$1.4 \cdot 10^{-17}$	0
invariant $q = Q_M(a)$, max rel. error	$2.0 \cdot 10^{-11}$	$6.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$6.8 \cdot 10^{-14}$

The bound stays within 4.5 to 6.3 times the realised displacement across this width range while both fall by more than a factor of 40. The normalized-readout control, in which the readout is $M^{-1/2}Uh$ with $O(1)$ entries trained at unit rate, removes the width dependence exactly: the encoder’s movement is the $M = 1$ value at every width, to 1.7×10^{-14} relative error. The eigenvalue constants $\lambda_\theta(0) = 1/4$ and $\lambda_\phi(0) = M/4$ hold for the mean-over-examples binary cross-entropy with a unit-scaled scalar logit; summing instead of averaging, or rescaling the logit, multiplies both blocks equally and leaves $\mathcal{D}_{\text{curv}}(0) = M$ unchanged.

Isotropic extension. Table 2 collects the verification of the general-initialization version, in which $a(0) = a_0$, $v(0) = v_0 \neq 0$ and the readouts start unequal but sum to exactly zero. The grid is $M \in \{1, 4, 16, 64, 256, 1024, 4096\}$, $a_0 \in \{-1.5, -0.3, 0, 0.4, 1.7\}$, $v_0 \in \{-1.3, -0.2, 0.05, 0.2, 1.0\}$ and $m \in \{2.0, 2.9, 4.0\}$: 525 non-trivial cases plus 54 immediate-stop cases with $a_0 \geq m$, integrated with DOP853 at $\text{rtol } 10^{-10}$ and $\text{atol } 10^{-12}$ with a terminal event at $q = m$.

The update fraction $(a(T_m) - a_0)/(m - a_0)$ falls towards zero in M at every (a_0, v_0) on the grid — for instance 0.288, 0.0423, 0.000964 at $M = 1, 16, 1024$ with $a_0 = -1.5$, $v_0 = 1$, $m = 2.9$ — while $a(T_m)/m$ and $a(T_m)/a_0$ converge to a_0/m and to 1 instead, so those two ratios are not

Table 2: Isotropic serial instance: verification summary and worst-case bound slacks over the whole grid, from `results/toy_serial_ce_isotropic_REPORT.md` §1–§2. Slacks are bound – value for upper bounds and value – bound for lower bounds; a negative entry would be a violation, and there are none. $\Delta = m - a_0$.

Quantity	Value
Max closed-form vs ODE relative error over $u, v(T_m), b(T_m), T_m$ (525 cases)	1.260×10^{-11}
Max relative error of the conserved quantity along the whole trajectory	1.518×10^{-10}
Max root accuracy $ Q_{\text{iso}}(u_M) - m $	3.553×10^{-15}
Max $ \sum_j \beta_j(0) $ (zero-sum initialization is bitwise exact)	0
Max spread of $\beta_j(T_m) - \beta_j(0)$	7.772×10^{-16}
Max frozen-comparator ODE vs closed-form relative error	5.993×10^{-11}
Max $M^{-1/2}$ -readout control vs its $M = 1$ value	1.335×10^{-12}
Bound violations, all cases and all bounds	0
Min slack, $u_M \leq \Delta/(1 + Mv_0^2)$	2.221×10^{-10}
Min slack, $u_M > 0$	4.333×10^{-5}
Min slack, $ v(T_m) - v_0 \leq \Delta^2/(2M v_0 ^3)$	1.493×10^{-9}
Min slack, encoder displacement bound	3.918×10^{-10}
Min slack, $T_m \leq \Psi_{a_0}(m)/(1 + Mv_0^2)$	1.748×10^{-9}
Min slack, $T_m \geq \Psi_{a_0}(m)/(1 + Mv_0^2 + 2\Delta^2/v_0^2)$	3.175×10^{-9}
Min slack, $q_J(t) \geq q_F(t)$	0
Min slack, $\sup_t(q_J - q_F) \leq C_0/(p_0 M v_0^2)$	1.495×10^{-4}
Reversal error at $\sigma = 1, m = 2, M = 64 / 1024 / 16384$	0.8207 / 0.8380 / 0.8397
Monte-Carlo limit $\Phi(m/2\sigma)$ on the same 4000 draws	0.8413
Draws disagreeing with the limit that satisfy $Mv_0^2 > m/(m - 2a_0)$	0 of 48000
Spectral-only comparator reversal error, every case	0

the quantities that vanish. Under the $M^{-1/2}$ readout the suppression is 0.336399 at every width, against 0.336, 0.0472 and 0.000969 for the unnormalized readout at $M = 1, 16$ and 1024. Every Monte-Carlo draw that falls short of the reversal limit fails the sufficient finite-width condition, either because $|v_0|$ is small or because a_0 sits just below $m/2$, where the threshold diverges.

C EXPERIMENT 2 DETAILS

C.1 GENERATOR

Images are 16×16 on a torus with $S = 256$ channels per pixel. Labels $y_p \in \{-1, +1\}$ are i.i.d. Rademacher. Fixed orthonormal directions $u, v_1, \dots, v_8 \in \mathbb{R}^S$ are drawn once per seed (QR of a Gaussian matrix). With $o_1, \dots, o_8$ the eight 3×3 offsets, the spectrum of pixel p is

$$x_p = \alpha y_p u + \beta \sum_{d=1}^8 (y_{p-o_d} + \tau \eta_{p,d}) v_d + \sigma \varepsilon_p, \quad \eta_{p,d}, \varepsilon_p \text{ standard Gaussian,}$$

so that the pixel at $p + o_d$ carries y_p along v_d : each pixel’s label is written into all eight neighbours, each along its own direction, while p ’s own v -coordinates carry the labels of p ’s neighbours, which are independent of y_p . Constants: $\alpha = 1.645$, $\beta = 5$, $\sigma = 1$, $\tau = 1.7023$; $K = 12$. The draw order is condition-independent, so the five conditions generated from one seed share y, η and ε : *iid*; *reversed* ($y_{p-o_d} \rightarrow -y_{p-o_d}$); *context-random* (an independent label field replaces y in the context term); *spectral-only* ($\beta = 0$); *context-only* ($\alpha = 0$). Training uses 256 images; each test condition 64 images; the spectral probe is fitted on 64 spectral-only images and evaluated on another 64. Seeds: directions $1000 + s$, training data $2000 + s$, test family $3000 + s$, probe $4000 + s$ and $5000 + s$, initialization s , for $s \in \{0, 1, 2\}$.

C.2 CALIBRATION AND READINESS (FIXED BEFORE ANY TRAINING RUN)

The oracle spectral accuracy is a Fisher discriminant on $u^\top x_p$; the oracle contextual accuracy is a Fisher discriminant on the 3×3 window of the eight projections $v_d^\top x$ (72 features), which contains

all eight copies of y_p . τ was set by bisection so that the contextual oracle equals 0.950 on seed-0 directions. Readiness is the same discriminant through a random encoder $W \sim \mathcal{N}(0, 1/S)^{12 \times 256}$: on the encoder output of the pixel itself for spectral-only data, and on the 3×3 window of encoder outputs for context-only data.

Table 3: Experiment 2 calibration (`results/exp2/calibration.json`). Oracle accuracies use the true directions; readiness uses a random encoder.

seed	oracle spectral	oracle context	random enc. spectral	random enc. context	random enc. both
0	0.951	0.950	0.647	0.910	0.915
1	0.951	0.949	0.653	0.914	0.918
2	0.948	0.950	0.608	0.905	0.906

The population optimum of the contextual cue alone is a cross-entropy of about 0.126 nats (spectral alone 0.127), so training losses below about 0.13 cannot be reached by either cue alone and the threshold 0.15 sits just above the contextual optimum.

C.3 MODEL, TRAINING AND ARMS

Encoder $W \in \mathbb{R}^{12 \times 256}$, no bias, $\mathcal{N}(0, 1/S)$ initialization. Head: 3×3 convolution to width M with bias, ReLU, 3×3 convolution to one logit without bias, circular padding, PyTorch default initialization; implemented as explicit matrix products with torus rolls (identical to the convolution to 10^{-15} in double precision). Loss: mean over the 65,536 training pixels of $\log(1 + e^{-y_p F_p})$. Full-batch gradient descent, $\eta = 10^{-3}$ for every parameter (the largest of $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$ with a monotone loss over 300 steps at $M = 2048$), no momentum, clipping or decay; at most 40,000 steps, stopping at loss < 0.10 . Snapshots at the first step with loss below each of 0.6, 0.5, 0.4, 0.3, 0.2, 0.15, 0.10; $L^* = 0.30$ is the pre-registered threshold. Gradient norms at a snapshot are those of the snapshotted iterate. Arms: `sp` (standard parameterization) and `mup` (readout $M^{-1/2}\gamma$, $\gamma = \sqrt{M} \times \text{default}$, same function at initialization; under plain gradient descent equal to a readout-only rate $1/M$) at $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; `ctxfree` (training context from an independent label field) and `frozen` (encoder fixed at initialization) at $M \in \{2, 8, 128, 512\}$; `lrmult` (readout rate $\times \{1/16, 1/4, 1, 4, 16\}$) and `headlr` (whole-head rate $\kappa \in \{1, 1/16, 1/256\}$, extended to $1/4096$ and $1/32768$ on seed 0 for the reversal outcome) at $M = 32$. Three seeds.

C.4 PRE-REGISTRATION AND AMENDMENTS

The plan (`paper/REFOCUS_PLAN_2026-09-10.md`, §4) fixed the generator, constants, arms, L^* , measures and predictions P1–P5 before the grid. Amendments recorded before the corresponding runs: widths 2 and 4 and fractional readout multipliers were added after a one-seed pilot at $M \in \{8, 2048\}$ showed complete suppression at L^* (§4.4a); the whole-head arm, the exact h_u export, paired test conditions and same-state gradient norms were added after the design review (§4.4b); the κ bracket was extended downward after the seed-0 pilot showed that the probe responds to κ while reversal accuracy does not within $\{1, 1/16, 1/256\}$ (§4.4c). Original predictions are reported as registered; added widths and multipliers are identified as amendments.

C.5 RESULTS

Tables generated by `paper/figures_src/make_tables_exp2.py` from `results/exp2_summary.csv`; the registered verdicts are the per-seed Spearman correlations listed at the end.

Runs that did not reach L^* . The following runs ended at the step budget above L^* : `headlr` $M = 32$, $\kappa = 0.000244141$, seed 0 (final loss 0.320 at step 40000); `headlr` $M = 32$, $\kappa = 3.05176e - 05$, seed 0 (final loss 0.347 at step 40000).

Registered verdicts. Spearman correlations across widths at L^* , per seed (P1–P3 as registered use a_u and reversal accuracy): `sp` seed 0: $\rho(a_u, M) = -0.54$, $\rho(\text{probe}, M) = -0.54$, $\rho(\text{rev}, M) =$

Table 4: Experiment 2 at the pre-registered threshold $L^* = 0.30$: mean $\pm$ sd over seeds (number of seeds). Probe gain is the discriminant accuracy on the encoder output minus its initial value (0.647, 0.653, 0.608 for seeds 0–2); $h_u = \|P_{\text{row}(W)}u\|^2$ (0.055 at initialization).

arm	M	probe gain	h_u	reversed acc.	context-random acc.	spectral-only acc.
sp	2	0.042 $\pm$ 0.028 (3)	0.086 $\pm$ 0.037 (3)	0.134 $\pm$ 0.006 (3)	0.503 $\pm$ 0.005 (3)	0.525 $\pm$ 0.016 (3)
sp	4	0.021 $\pm$ 0.010 (3)	0.065 $\pm$ 0.028 (3)	0.125 $\pm$ 0.008 (3)	0.508 $\pm$ 0.007 (3)	0.525 $\pm$ 0.021 (3)
sp	8	0.026 $\pm$ 0.016 (3)	0.067 $\pm$ 0.021 (3)	0.124 $\pm$ 0.002 (3)	0.507 $\pm$ 0.005 (3)	0.524 $\pm$ 0.017 (3)
sp	32	0.022 $\pm$ 0.007 (3)	0.067 $\pm$ 0.027 (3)	0.121 $\pm$ 0.008 (3)	0.508 $\pm$ 0.008 (3)	0.531 $\pm$ 0.022 (3)
sp	128	0.009 $\pm$ 0.006 (3)	0.055 $\pm$ 0.022 (3)	0.117 $\pm$ 0.012 (3)	0.507 $\pm$ 0.008 (3)	0.524 $\pm$ 0.019 (3)
sp	512	0.005 $\pm$ 0.001 (3)	0.051 $\pm$ 0.018 (3)	0.115 $\pm$ 0.007 (3)	0.508 $\pm$ 0.006 (3)	0.525 $\pm$ 0.022 (3)
sp	2048	0.002 $\pm$ 0.000 (3)	0.049 $\pm$ 0.017 (3)	0.117 $\pm$ 0.005 (3)	0.507 $\pm$ 0.006 (3)	0.522 $\pm$ 0.017 (3)
mup	2	0.054 $\pm$ 0.034 (3)	0.098 $\pm$ 0.042 (3)	0.134 $\pm$ 0.006 (3)	0.504 $\pm$ 0.004 (3)	0.528 $\pm$ 0.015 (3)
mup	4	0.036 $\pm$ 0.019 (3)	0.079 $\pm$ 0.037 (3)	0.126 $\pm$ 0.011 (3)	0.507 $\pm$ 0.007 (3)	0.530 $\pm$ 0.026 (3)
mup	8	0.039 $\pm$ 0.023 (3)	0.080 $\pm$ 0.034 (3)	0.120 $\pm$ 0.002 (3)	0.506 $\pm$ 0.006 (3)	0.530 $\pm$ 0.024 (3)
mup	32	0.044 $\pm$ 0.014 (3)	0.088 $\pm$ 0.037 (3)	0.115 $\pm$ 0.009 (3)	0.507 $\pm$ 0.007 (3)	0.539 $\pm$ 0.026 (3)
mup	128	0.026 $\pm$ 0.016 (3)	0.070 $\pm$ 0.031 (3)	0.108 $\pm$ 0.012 (3)	0.505 $\pm$ 0.008 (3)	0.526 $\pm$ 0.018 (3)
mup	512	0.026 $\pm$ 0.019 (3)	0.071 $\pm$ 0.034 (3)	0.112 $\pm$ 0.008 (3)	0.505 $\pm$ 0.007 (3)	0.526 $\pm$ 0.022 (3)
mup	2048	0.028 $\pm$ 0.004 (3)	0.070 $\pm$ 0.018 (3)	0.104 $\pm$ 0.005 (3)	0.504 $\pm$ 0.006 (3)	0.523 $\pm$ 0.010 (3)
ctxfree	2	0.296 $\pm$ 0.019 (3)	0.863 $\pm$ 0.048 (3)	0.877 $\pm$ 0.002 (3)	0.876 $\pm$ 0.003 (3)	0.877 $\pm$ 0.002 (3)
ctxfree	8	0.281 $\pm$ 0.022 (3)	0.741 $\pm$ 0.014 (3)	0.893 $\pm$ 0.001 (3)	0.893 $\pm$ 0.001 (3)	0.899 $\pm$ 0.002 (3)
ctxfree	128	0.268 $\pm$ 0.023 (3)	0.650 $\pm$ 0.014 (3)	0.896 $\pm$ 0.001 (3)	0.895 $\pm$ 0.002 (3)	0.903 $\pm$ 0.004 (3)
ctxfree	512	0.253 $\pm$ 0.025 (3)	0.564 $\pm$ 0.005 (3)	0.886 $\pm$ 0.002 (3)	0.887 $\pm$ 0.004 (3)	0.891 $\pm$ 0.004 (3)
frozen	2	0.000 $\pm$ 0.000 (3)	0.048 $\pm$ 0.017 (3)	0.139 $\pm$ 0.009 (3)	0.508 $\pm$ 0.006 (3)	0.521 $\pm$ 0.018 (3)
frozen	8	0.000 $\pm$ 0.000 (3)	0.048 $\pm$ 0.017 (3)	0.134 $\pm$ 0.006 (3)	0.511 $\pm$ 0.007 (3)	0.524 $\pm$ 0.019 (3)
frozen	128	0.000 $\pm$ 0.000 (3)	0.048 $\pm$ 0.017 (3)	0.127 $\pm$ 0.008 (3)	0.508 $\pm$ 0.008 (3)	0.524 $\pm$ 0.021 (3)
frozen	512	0.000 $\pm$ 0.000 (3)	0.048 $\pm$ 0.017 (3)	0.120 $\pm$ 0.006 (3)	0.508 $\pm$ 0.008 (3)	0.526 $\pm$ 0.019 (3)

Table 5: Experiment 2 at the secondary threshold 0.15 (labelled; not pre-registered): mean $\pm$ sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
sp	2	0.219 $\pm$ 0.039 (3)	0.462 $\pm$ 0.232 (3)	0.131 $\pm$ 0.088 (3)	0.543 $\pm$ 0.046 (3)
sp	4	0.094 $\pm$ 0.019 (3)	0.149 $\pm$ 0.056 (3)	0.070 $\pm$ 0.006 (3)	0.506 $\pm$ 0.007 (3)
sp	8	0.121 $\pm$ 0.020 (3)	0.184 $\pm$ 0.046 (3)	0.073 $\pm$ 0.007 (3)	0.508 $\pm$ 0.008 (3)
sp	32	0.094 $\pm$ 0.014 (3)	0.147 $\pm$ 0.051 (3)	0.072 $\pm$ 0.008 (3)	0.509 $\pm$ 0.009 (3)
sp	128	0.060 $\pm$ 0.022 (3)	0.106 $\pm$ 0.052 (3)	0.070 $\pm$ 0.009 (3)	0.507 $\pm$ 0.009 (3)
sp	512	0.041 $\pm$ 0.012 (3)	0.084 $\pm$ 0.036 (3)	0.071 $\pm$ 0.011 (3)	0.508 $\pm$ 0.009 (3)
sp	2048	0.032 $\pm$ 0.005 (3)	0.073 $\pm$ 0.026 (3)	0.076 $\pm$ 0.010 (3)	0.509 $\pm$ 0.010 (3)
mup	2	0.230 $\pm$ 0.032 (3)	0.499 $\pm$ 0.216 (3)	0.136 $\pm$ 0.093 (3)	0.546 $\pm$ 0.051 (3)
mup	4	0.142 $\pm$ 0.020 (3)	0.233 $\pm$ 0.079 (3)	0.074 $\pm$ 0.008 (3)	0.508 $\pm$ 0.006 (3)
mup	8	0.148 $\pm$ 0.030 (3)	0.241 $\pm$ 0.090 (3)	0.073 $\pm$ 0.009 (3)	0.509 $\pm$ 0.008 (3)
mup	32	0.134 $\pm$ 0.015 (3)	0.213 $\pm$ 0.068 (3)	0.070 $\pm$ 0.008 (3)	0.507 $\pm$ 0.008 (3)
mup	128	0.088 $\pm$ 0.033 (3)	0.146 $\pm$ 0.070 (3)	0.064 $\pm$ 0.006 (3)	0.505 $\pm$ 0.008 (3)
mup	512	0.108 $\pm$ 0.041 (3)	0.178 $\pm$ 0.104 (3)	0.067 $\pm$ 0.009 (3)	0.506 $\pm$ 0.007 (3)
mup	2048	0.100 $\pm$ 0.006 (3)	0.153 $\pm$ 0.035 (3)	0.065 $\pm$ 0.003 (3)	0.506 $\pm$ 0.007 (3)
ctxfree	2	0.317 $\pm$ 0.024 (2)	0.990 $\pm$ 0.000 (2)	0.940 $\pm$ 0.006 (2)	0.940 $\pm$ 0.007 (2)
ctxfree	8	0.308 $\pm$ 0.023 (3)	0.977 $\pm$ 0.005 (3)	0.937 $\pm$ 0.002 (3)	0.938 $\pm$ 0.002 (3)
ctxfree	128	0.304 $\pm$ 0.023 (3)	0.937 $\pm$ 0.002 (3)	0.941 $\pm$ 0.000 (3)	0.941 $\pm$ 0.000 (3)
ctxfree	512	0.302 $\pm$ 0.023 (3)	0.913 $\pm$ 0.004 (3)	0.940 $\pm$ 0.001 (3)	0.940 $\pm$ 0.000 (3)

Table 6: Experiment 2 at the secondary threshold 0.1 (labelled; not pre-registered): mean $\pm$ sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
sp	2	0.274 $\pm$ 0.021 (2)	0.629 $\pm$ 0.021 (2)	0.137 $\pm$ 0.006 (2)	0.555 $\pm$ 0.008 (2)
sp	4	0.227 $\pm$ 0.006 (3)	0.462 $\pm$ 0.055 (3)	0.089 $\pm$ 0.009 (3)	0.525 $\pm$ 0.007 (3)
sp	8	0.241 $\pm$ 0.024 (3)	0.514 $\pm$ 0.028 (3)	0.099 $\pm$ 0.008 (3)	0.531 $\pm$ 0.008 (3)
sp	32	0.227 $\pm$ 0.016 (3)	0.459 $\pm$ 0.044 (3)	0.098 $\pm$ 0.008 (3)	0.530 $\pm$ 0.007 (3)
sp	128	0.206 $\pm$ 0.011 (3)	0.383 $\pm$ 0.044 (3)	0.096 $\pm$ 0.010 (3)	0.528 $\pm$ 0.009 (3)
sp	512	0.170 $\pm$ 0.006 (3)	0.282 $\pm$ 0.047 (3)	0.096 $\pm$ 0.010 (3)	0.528 $\pm$ 0.009 (3)
sp	2048	0.142 $\pm$ 0.007 (3)	0.218 $\pm$ 0.038 (3)	0.106 $\pm$ 0.013 (3)	0.533 $\pm$ 0.011 (3)
mup	2	0.281 $\pm$ 0.023 (2)	0.669 $\pm$ 0.017 (2)	0.138 $\pm$ 0.003 (2)	0.555 $\pm$ 0.006 (2)
mup	4	0.250 $\pm$ 0.008 (3)	0.566 $\pm$ 0.063 (3)	0.095 $\pm$ 0.011 (3)	0.529 $\pm$ 0.008 (3)
mup	8	0.248 $\pm$ 0.003 (2)	0.618 $\pm$ 0.051 (2)	0.102 $\pm$ 0.014 (2)	0.536 $\pm$ 0.009 (2)
mup	32	0.243 $\pm$ 0.002 (2)	0.591 $\pm$ 0.024 (2)	0.089 $\pm$ 0.000 (2)	0.528 $\pm$ 0.001 (2)
mup	128	0.237 (1)	0.579 (1)	0.084 (1)	0.524 (1)
mup	512	0.242 (1)	0.611 (1)	0.086 (1)	0.525 (1)

Table 7: Whole-head rate multiplier κ at $M = 32$ (arm `headlr`) at L^* ; rows marked $\dagger$ never reached L^* within 40,000 steps and show the final state (loss in parentheses).

seed	κ	step	probe gain	h_u	reversed	context-random	spectral-only
0	3.05176e-05 $\dagger$	40000 (0.347)	0.267	0.701	0.208	0.528	0.689
0	0.000244141 $\dagger$	40000 (0.320)	0.266	0.693	0.184	0.529	0.697
0	0.00390625	22708	0.223	0.475	0.152	0.519	0.641
0	0.0625	7611	0.106	0.177	0.129	0.513	0.568
0	1	1671	0.028	0.079	0.128	0.515	0.536
1	0.00390625	34174	0.203	0.443	0.146	0.510	0.615
1	0.0625	8977	0.100	0.181	0.129	0.509	0.575
1	1	1580	0.025	0.085	0.121	0.511	0.551
2	0.00390625	32068	0.205	0.301	0.130	0.500	0.570
2	0.0625	8704	0.056	0.067	0.110	0.498	0.520
2	1	1881	0.014	0.036	0.112	0.500	0.507

Table 8: Readout learning-rate multiplier at $M = 32$ (arm `lrmult`) at L^* : mean $\pm$ sd over three seeds.

multiplier	probe gain	h_u	reversed acc.	context-random acc.
0.0625	0.041 $\pm$ 0.013	0.085 $\pm$ 0.036	0.115 $\pm$ 0.009	0.507 $\pm$ 0.007
0.25	0.031 $\pm$ 0.012	0.076 $\pm$ 0.033	0.116 $\pm$ 0.009	0.507 $\pm$ 0.007
1	0.022 $\pm$ 0.007	0.067 $\pm$ 0.027	0.121 $\pm$ 0.008	0.508 $\pm$ 0.008
4	0.015 $\pm$ 0.003	0.059 $\pm$ 0.022	0.127 $\pm$ 0.008	0.509 $\pm$ 0.008
16	0.008 $\pm$ 0.001	0.054 $\pm$ 0.019	0.133 $\pm$ 0.009	0.510 $\pm$ 0.006

-0.64 over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; sp seed 1: $\rho(a_u, M) = -1.00$, $\rho(\text{probe}, M) = -1.00$, $\rho(\text{rev}, M) = -0.96$ over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; sp seed 2: $\rho(a_u, M) = -0.96$, $\rho(\text{probe}, M) = -0.93$, $\rho(\text{rev}, M) = -0.82$ over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; mup seed 0: $\rho(a_u, M) = +0.25$, $\rho(\text{probe}, M) = -0.04$, $\rho(\text{rev}, M) = -0.89$ over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; mup seed 1: $\rho(a_u, M) = -0.96$, $\rho(\text{probe}, M) = -0.96$, $\rho(\text{rev}, M) = -0.89$ over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; mup seed 2: $\rho(a_u, M) = -0.54$, $\rho(\text{probe}, M) = -0.46$, $\rho(\text{rev}, M) = -0.89$ over $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; ctxfree seed 0: $\rho(a_u, M) = -1.00$, $\rho(\text{probe}, M) = -1.00$, $\rho(\text{rev}, M) = +0.40$ over $M \in \{2, 8, 128, 512\}$; ctxfree seed 1: $\rho(a_u, M) = -1.00$, $\rho(\text{probe}, M) = -1.00$, $\rho(\text{rev}, M) = +0.40$ over $M \in \{2, 8, 128, 512\}$; ctxfree seed 2: $\rho(a_u, M) = -1.00$, $\rho(\text{probe}, M) = -1.00$, $\rho(\text{rev}, M) = +0.40$ over $M \in \{2, 8, 128, 512\}$; frozen seed 0: $\rho(a_u, M) = +nan$, $\rho(\text{probe}, M) = +nan$, $\rho(\text{rev}, M) = -1.00$ over $M \in \{2, 8, 128, 512\}$; frozen seed 1: $\rho(a_u, M) = +nan$, $\rho(\text{probe}, M) = +nan$, $\rho(\text{rev}, M) = -1.00$ over $M \in \{2, 8, 128, 512\}$; frozen seed 2: $\rho(a_u, M) = +nan$, $\rho(\text{probe}, M) = +nan$, $\rho(\text{rev}, M) = -1.00$ over $M \in \{2, 8, 128, 512\}$. P2's second clause (a_u at $M = 2048$, normalized readout versus standard): seed 0: 0.0059 vs 0.0045; seed 1: 0.0062 vs 0.0045; seed 2: 0.0043 vs 0.0027.

D EXPERIMENT 3 DETAILS

D.1 DATA

Breast-tissue QCL micro-array cores, fold 0 of the five-fold patient-level split used throughout this work (115 training, 28 validation, 26 test cores). Each labelled tissue pixel contributes the 942-dimensional vector of absorbance, first and second derivative (3×314 channels). Up to 600 (training) or 300 (validation, test) pixels per class per core were sampled with a fixed seed: 75,297 training pixels from 79 cores, 11,024 validation from 20, 8,038 test from 16, over the four classes; the experiment uses the pair Cancer Epithelium (+1; 29,603 training pixels) versus Cancer-Associated Stroma (-1 ; 21,923).

D.2 READINESS SCAN

For each class pair, bottleneck K and preprocessing, Table 9 reports the centre-only oracle (ridge Fisher discriminant on the preprocessed 942-vector, fitted on one half of the training cores and evaluated on the other half) and the same discriminant through three random encoders $W \sim$

$\mathcal{N}(0, 1/942)^{K \times 942}$. Standardization is per feature; whitening is PCA whitening with a relative ridge of 10^{-3} on the eigenvalues; both are estimated on the fitting half.

Table 9: Readiness scan (results/exp3/readiness_scan.csv): centre-only oracle and random-encoder probe (mean over three encoders) by pair, K and preprocessing. $|\cos|$ is the cosine between the class-mean contrast and the first principal direction of the standardized training half.

pair	preprocessing	oracle	$K = 2$	$K = 4$	$K = 12$	$K = 32$	$ \cos $
CancerEpi vs CAS	standardized	0.969	0.726	0.803	0.955	0.968	0.50
CancerEpi vs CAS	whitened	0.944	0.503	0.486	0.539	0.591	0.50
NormalStroma vs CAS	standardized	0.810	0.580	0.639	0.674	0.741	0.44
NormalStroma vs CAS	whitened	0.768	0.502	0.509	0.529	0.600	0.44
NormalEpi vs CancerEpi	standardized	0.758	0.547	0.573	0.662	0.724	0.20
NormalEpi vs CancerEpi	whitened	0.761	0.575	0.545	0.624	0.580	0.20
NormalStroma vs CancerEpi	standardized	0.989	0.781	0.874	0.978	0.990	0.88
NormalStroma vs CancerEpi	whitened	0.974	0.579	0.541	0.549	0.626	0.88

With standardized inputs a random encoder already reads the epithelium-versus-stroma contrasts, because they lie along the top principal directions; whitening removes that accessibility while the oracle stays high. The two regimes of Section 5 follow: the same pair, $K = 12$, standardized (*ready*) or whitened (*unready*).

D.3 PATCH CONSTRUCTION

Each training example is a real centre pixel with its recorded label and eight real donor spectra sampled uniformly from the same split side (resampled once if the donor’s core equals the centre’s), independent of the centre label. In the preprocessed space, donor d receives the constructed cue $\gamma_d(c s + \tau \eta_d) e_d$, where $e_1, \dots, e_8$ are the first eight principal coordinates (in the standardized regime the directions $v_1, \dots, v_8$ with $\gamma_d = 3\sqrt{\lambda_d}$; in the whitened regime the first eight whitened coordinates with $\gamma_d = 30$), c is the centre’s label code, and $s = +1$ (informative), $s = -1$ (reversed) or c replaced by an independent random label (context-random); *spectral-only* removes the cue and keeps the donors. The draw order is condition-independent, so the four conditions generated from one seed share centres, donors and η . Preprocessing, principal directions and λ_d are estimated on training pixels only. Training: 24,000 class-balanced centres; probe fitting: 6,000 disjoint training pixels; evaluation: up to 6,000 validation and 6,000 test centres with donors from their own side. τ is calibrated per regime on training-core halves so that the neighbourhood oracle (a discriminant on the eight cue projections) matches the centre-only oracle.

Table 10: Experiment 3 calibration (results/exp3/calibration3.json). Readiness: discriminant through a random 12×942 encoder on the centre alone and on the encoded 3×3 patch (three encoders).

regime	τ	centre oracle (fit/val)	context oracle	random enc. centre	random enc. patch
ready (standardized)	1.475	0.969 / 0.959	0.969	0.935, 0.970, 0.952	0.988, 0.994, 0.991
unready (whitened)	1.735	0.948 / 0.884	0.948	0.476, 0.512, 0.605	0.948, 0.947, 0.950

D.4 MODEL, TRAINING AND ARMS

Encoder $W \in \mathbb{R}^{12 \times 942}$ applied to the nine spectra; head $h = \text{ReLU}(W_1 \text{vec}(z) + b_1)$ of width M over the $9K = 108$ encoded coordinates and a linear readout to one logit; margin loss; full-batch gradient descent with $\eta = 10^{-3}$, at most 40,000 steps, the same thresholds and snapshot rule as Experiment 2. Arms per regime: sp at $M \in \{8, 32, 128, 512\}$; headlr with $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$; ctxfree (context-random training) at $M \in \{8, 128\}$; frozen at $M = 32$; seeds 0–2. Spectral probe: ridge discriminant on the encoder output of centre pixels, fitted on the probe set and evaluated on validation (and test) centres, reported as the gain over its initial value. Reliance and failure are measured on validation patients with the paired conditions; test patients are reported separately and were not used for any choice.

D.5 RESULTS

Tables generated by `paper/figures_src/make_tables_exp3.py` from `results/exp3_summary.csv`.

Table 11: Experiment 3, unready (whitened, $\gamma = 30$) regime, at $L^* = 0.30$: mean $\pm$ sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
sp	8	126 $\pm$ 51	0.015 $\pm$ 0.031	0.083 $\pm$ 0.014	0.500 $\pm$ 0.012	0.485 $\pm$ 0.028	0.083 $\pm$ 0.005	0.496 $\pm$ 0.000 (3)
sp	32	113 $\pm$ 22	0.032 $\pm$ 0.042	0.072 $\pm$ 0.006	0.500 $\pm$ 0.010	0.510 $\pm$ 0.006	0.070 $\pm$ 0.003	0.495 $\pm$ 0.001 (3)
sp	128	100 $\pm$ 12	0.026 $\pm$ 0.017	0.075 $\pm$ 0.003	0.500 $\pm$ 0.010	0.495 $\pm$ 0.015	0.073 $\pm$ 0.006	0.496 $\pm$ 0.004 (3)
sp	512	69 $\pm$ 8	-0.012 $\pm$ 0.022	0.066 $\pm$ 0.001	0.498 $\pm$ 0.011	0.486 $\pm$ 0.010	0.067 $\pm$ 0.005	0.495 $\pm$ 0.004 (3)
headlr	32 / 0.00390625	185 $\pm$ 43	0.046 $\pm$ 0.052	0.075 $\pm$ 0.012	0.500 $\pm$ 0.013	0.512 $\pm$ 0.011	0.072 $\pm$ 0.002	0.494 $\pm$ 0.001 (3)
headlr	32 / 0.0625	177 $\pm$ 40	0.044 $\pm$ 0.051	0.074 $\pm$ 0.011	0.500 $\pm$ 0.013	0.512 $\pm$ 0.010	0.073 $\pm$ 0.001	0.495 $\pm$ 0.002 (3)
headlr	32	113 $\pm$ 22	0.032 $\pm$ 0.042	0.072 $\pm$ 0.006	0.500 $\pm$ 0.010	0.510 $\pm$ 0.006	0.070 $\pm$ 0.003	0.495 $\pm$ 0.001 (3)
ctxfree	8	16780 $\pm$ 2138	0.352 $\pm$ 0.008	0.925 $\pm$ 0.015	0.923 $\pm$ 0.015	0.924 $\pm$ 0.010	0.935 $\pm$ 0.023	0.934 $\pm$ 0.023 (3)
ctxfree	128	14954 $\pm$ 488	0.336 $\pm$ 0.014	0.911 $\pm$ 0.030	0.908 $\pm$ 0.029	0.926 $\pm$ 0.028	0.930 $\pm$ 0.014	0.928 $\pm$ 0.009 (3)
frozen	32	395 $\pm$ 95	0.000 $\pm$ 0.000	0.076 $\pm$ 0.006	0.500 $\pm$ 0.009	0.515 $\pm$ 0.008	0.076 $\pm$ 0.002	0.496 $\pm$ 0.002 (3)

Table 12: Experiment 3, unready (whitened, $\gamma = 10$; amendment) regime, at $L^* = 0.30$: mean $\pm$ sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
sp	32	592 $\pm$ 70	0.076 $\pm$ 0.055	0.067 $\pm$ 0.004	0.501 $\pm$ 0.011	0.512 $\pm$ 0.007	0.063 $\pm$ 0.002	0.496 $\pm$ 0.002 (3)
headlr	32 / 0.00390625	1579 $\pm$ 318	0.134 $\pm$ 0.072	0.066 $\pm$ 0.010	0.501 $\pm$ 0.011	0.508 $\pm$ 0.015	0.065 $\pm$ 0.005	0.496 $\pm$ 0.001 (3)
headlr	32 / 0.0625	1337 $\pm$ 227	0.124 $\pm$ 0.068	0.065 $\pm$ 0.009	0.501 $\pm$ 0.011	0.508 $\pm$ 0.013	0.065 $\pm$ 0.003	0.496 $\pm$ 0.001 (3)
headlr	32	592 $\pm$ 70	0.076 $\pm$ 0.055	0.067 $\pm$ 0.004	0.501 $\pm$ 0.011	0.512 $\pm$ 0.007	0.063 $\pm$ 0.002	0.496 $\pm$ 0.002 (3)
ctxfree	32	16003 $\pm$ 231	0.350 $\pm$ 0.012	0.916 $\pm$ 0.005	0.916 $\pm$ 0.006	0.925 $\pm$ 0.009	0.923 $\pm$ 0.011	0.924 $\pm$ 0.011 (3)
frozen	32	2006 $\pm$ 310	0.000 $\pm$ 0.000	0.080 $\pm$ 0.008	0.502 $\pm$ 0.011	0.511 $\pm$ 0.007	0.073 $\pm$ 0.004	0.495 $\pm$ 0.001 (3)

Table 13: Experiment 3, ready (standardized) regime, at $L^* = 0.30$: mean $\pm$ sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
sp	8	132 $\pm$ 59	0.000 $\pm$ 0.017	0.087 $\pm$ 0.004	0.502 $\pm$ 0.017	0.522 $\pm$ 0.035	0.088 $\pm$ 0.009	0.499 $\pm$ 0.003 (3)
sp	32	118 $\pm$ 22	0.002 $\pm$ 0.013	0.083 $\pm$ 0.013	0.502 $\pm$ 0.013	0.526 $\pm$ 0.026	0.088 $\pm$ 0.013	0.500 $\pm$ 0.003 (3)
sp	128	103 $\pm$ 20	0.006 $\pm$ 0.016	0.102 $\pm$ 0.004	0.512 $\pm$ 0.011	0.578 $\pm$ 0.060	0.102 $\pm$ 0.009	0.511 $\pm$ 0.008 (3)
sp	512	70 $\pm$ 3	0.002 $\pm$ 0.014	0.092 $\pm$ 0.012	0.508 $\pm$ 0.011	0.544 $\pm$ 0.017	0.092 $\pm$ 0.021	0.504 $\pm$ 0.005 (3)
headlr	32 / 0.00390625	193 $\pm$ 31	0.003 $\pm$ 0.013	0.083 $\pm$ 0.010	0.503 $\pm$ 0.013	0.525 $\pm$ 0.028	0.086 $\pm$ 0.008	0.501 $\pm$ 0.001 (3)
headlr	32 / 0.0625	184 $\pm$ 29	0.002 $\pm$ 0.013	0.084 $\pm$ 0.011	0.503 $\pm$ 0.013	0.524 $\pm$ 0.028	0.086 $\pm$ 0.008	0.501 $\pm$ 0.001 (3)
headlr	32	118 $\pm$ 22	0.002 $\pm$ 0.013	0.083 $\pm$ 0.013	0.502 $\pm$ 0.013	0.526 $\pm$ 0.026	0.088 $\pm$ 0.013	0.500 $\pm$ 0.003 (3)
ctxfree	8	1128 $\pm$ 116	0.015 $\pm$ 0.021	0.929 $\pm$ 0.002	0.923 $\pm$ 0.001	0.931 $\pm$ 0.005	0.953 $\pm$ 0.008	0.947 $\pm$ 0.004 (3)
ctxfree	128	841 $\pm$ 112	0.012 $\pm$ 0.021	0.927 $\pm$ 0.006	0.925 $\pm$ 0.006	0.929 $\pm$ 0.004	0.955 $\pm$ 0.001	0.954 $\pm$ 0.000 (3)
frozen	32	442 $\pm$ 114	0.000 $\pm$ 0.000	0.091 $\pm$ 0.022	0.504 $\pm$ 0.013	0.529 $\pm$ 0.021	0.093 $\pm$ 0.027	0.499 $\pm$ 0.006 (3)

Runs that did not reach L^* . Every run reached L^* .

E SCOPE LIMITATIONS FROM A PRODUCTION MODEL

This appendix reproduces, from the earlier diagnostic draft of this work, the measurements on a production spectral-spatial segmentation model that are summarized in Section 6. They concern

initialization curvature and a controlled retraining study; they do not measure the mechanism of Theorem 1 and are included to document why scalar curvature is not the diagnostic used in this paper and why the earlier apparent freezing advantage is not evidence for or against it. The data are breast-tissue QCL micro-array cores with a prostate QCL replication. Theorem and proposition numbers quoted inside this appendix refer to the earlier draft’s framework and are retained only where the sentence is a scope statement about that framework.

E.1 PRODUCTION MODEL DETAILS

E.1.1 PIPELINE, CAPACITIES AND DATA

The data are 336×336 pixel tissue-microarray cores collected by quantum-cascade-laser infrared spectroscopy over the $1000\text{--}1800\text{ cm}^{-1}$ fingerprint region, with per-pixel four-class tissue labels. Each pixel carries an $S = 942$ dimensional vector, obtained by stacking baseline-corrected absorbance with its first two derivatives. The compositional model is the Block Vision Transformer of O’Leary et al. (2026): a linear per-pixel $f_\theta : \mathbb{R}^{942} \rightarrow \mathbb{R}^K$, then 16×16 patch tokenization, a 12-layer, 12-head vision transformer of embedding width M , a transposed-convolution decoder and a convolutional segmentation head. Parameter counts in Table 14 are read from the instantiated model.

Table 14: Module capacities of the production pipeline, counted from the instantiated model. K is the spectral bottleneck width. The last row is the single-channel field-standard input convention, for comparison.

Configuration	C_f	C_g	C_g/C_f
$K = 64$, three-channel input	61,108	18,271,540	299
$K = 128$, three-channel input	121,588	21,472,564	177
$K = 64$, single-channel input	20,916	18,271,540	874

Channel convention. The published architecture operates on the second-derivative spectrum alone, the field-standard preprocessing in infrared histopathology. The models analysed here instead stack raw absorbance with its first two derivatives, following a companion tokenization study (anonymous, under review). The choice affects only C_f , and the three-channel convention is the conservative one for the present purpose: under the field standard the capacity asymmetry is roughly three times larger. Within the spatial module the dominant blocks are the transformer (5.34×10^6 parameters) and the transposed convolution (9.44×10^6), both $\Theta(M^2)$ in the embedding width.

The head hypothesis is structurally unmet. The segmentation head is the chain $\text{Conv}_{3 \times 3}(M+K \rightarrow 96) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3}(96 \rightarrow 48) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3}(48 \rightarrow 4)$, so the final affine classifier sees $48 \times 3 \times 3 = 432$ input features at every width we build ($M \in \{48, 192, 384\}$, $K \in \{64, 128\}$): the $\Theta(M)$ feature map is separated from the logits by two nonlinearities. The curvature-disparity theorem requires a dense classifier over penultimate features of dimension $\Theta(M)$ with no downstream nonlinearity, and that hypothesis is not satisfied here. Failing a hypothesis removes a guarantee; it does not reverse the conclusion.

E.1.2 GEOMETRY AT INITIALIZATION

Gauss–Newton blocks of the instantiated pipeline were measured on real training batches at random initialization, by Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation, sweeping $M \in \{48, 96, 192, 384\}$ with three seeds and four batches per width. Table 15 collects the results.

Two controls bracket the denominator. Removing the spectral normalization deflates $\lambda_{\max}(G_{\theta\theta})$ by $2.4 \times$ ($471 \rightarrow 194$ at $M = 192$) and pushes the ratio across parity (1.18 at $M = 192$, 1.31 at $M = 384$). Input centering changes nothing when the architecture contains internal batch normalization: $\mathcal{D}_{\text{curv}}$ is identical to three decimals with and without centering, because a linear projection followed by normalization absorbs constant input shifts exactly. The near-rank-one input statistics make the cap side of the curvature-disparity bound uninformative at buildable widths independently of the

Table 15: Production geometry at initialization. Ratios are mean $\pm$ sd over three seeds and four batches per width. Slopes are log-log fits over the width sweep. Effective ranks are out of $S = 942$. Directional entries are measured at $M = 192$ over the nine measurements whose batch contains both classes (three seeds $\times$ three batches).

Quantity	Breast	Prostate
$\mathcal{D}_{\text{curv}}$ at $M = 48$	0.33 ± 0.12	—
$\mathcal{D}_{\text{curv}}$ at $M = 192$	0.46 ± 0.21	—
$\mathcal{D}_{\text{curv}}$ at $M = 384$	0.70 ± 0.19	1.09 ± 0.52
log-log slope of $\lambda_{\max}(G_{\theta\theta})$ vs M	-0.003	-0.012
log-log slope of $\lambda_{\max}(G_{\phi\phi})$ vs M	$0.38 (115 \rightarrow 258)$	0.45
Effective rank of Σ_X , raw inputs	1.07	1.02
Effective rank of Σ_X , post-normalization	1.21	—
$ \cos $ between v_1 and the mean spectrum (post-norm./raw)	$> 0.995 / > 0.9999$	—
θ -block top eigenvector’s norm fraction in the v_1 -induced subspace	0.954 ± 0.017	—
Trace fraction in the top 40 of 61,108 spectral directions	0.83 ± 0.12	—
Curvature ratio, v_1 vs full class contrast ($ \cos(d, v_1) = 0.45$)	$3.7 \times (2.6-4.8)$	—
Curvature ratio, v_1 vs v_1 -orthogonalized contrast	$12-28 \times$	—
Squared-norm fraction of the contrast retained after orthogonalization	$76-84\%$	—
Rank at which $\lambda_{\max}(G_{\phi\phi})$ overtakes the θ spectrum	$2-5$	—

head-hypothesis failure: the data concentrate almost all their second-moment mass in one direction, and that direction is the mean spectrum. The directional measurements establish an anisotropy, not its consequence for learning; whether it starves learning is a hypothesis, not a measurement.

E.1.3 WHAT CARRIES THE WIDTH TREND

The measurements in this subsection are made on the production model at random initialization, at three widths $M \in \{48, 192, 384\}$ and three seeds, on one core (the densest of the first 24 breast fold-0 training cores, with 29,664 labelled pixels of 112,896), with cuDNN TF32 disabled. Four normalization modes are compared: `train` (both head batch normalizations using batch statistics), `eval_bn1` and `eval_bn2` (only the first or only the second head normalization at its initial running statistics) and `eval_head` (both). Artifacts: `results/exp1_8c_interior_witness.csv` and `results/exp1_8d_REPORT.md`.

The incoming features are flat in width. The only downstream layer whose input dimension grows with M is the first head convolution. The top eigenvalue of the patch-unfolded second-moment matrix of the features entering it is unchanged to four significant figures across the three widths, at 356.3, 403.7 and 332.4 for seeds 0, 1 and 2 respectively; the leading direction sits inside the width-independent $K = 64$ skip channels. Table 16 also records a discrepancy that matters for any normalization argument: the same Gram computed over the full batch-normalization group (all positions, ignored pixels included) rather than over labelled pixels is substantially smaller, and the cosine between the two centered top eigenvectors is far from one.

Table 16: Incoming 3×3 -patch Gram of the first head convolution, top eigenvalue, from `results/exp1_8d_REPORT.md`. Values are identical across $M \in \{48, 192, 384\}$ and across all four normalization modes, so a single column suffices. *labelled* uses the labelled-pixel mask; *full group* uses the batch-normalization group. The last column is the cosine between the two centered top eigenvectors.

seed	labelled	labelled, centered	full group	cos(labelled, full)
0	356.3	68.19	263.3	0.2793
1	403.7	54.60	306.8	0.5502
2	332.4	67.14	253.0	0.2176

Normalization quantities. Table 17 gives the per-channel batch statistics of the two head normalizations, measured over the group they actually normalize, together with the activation energy

entering and leaving the first one. Under fan-in initialization the pre-normalization variance of the first head normalization falls roughly as $1/(M + K)$, and its inverse-standard-deviation gain grows correspondingly. The stabilizer $\varepsilon = 10^{-5}$ is never within three orders of magnitude of the variance: the fraction of channels with variance below 10ε is 0 in every row of both normalizations at every width, seed and mode.

Table 17: Head normalization quantities at initialization, `train` mode, from `results/exp1_8d_REPORT.md` Table 1. Per-seed values are per-channel medians (variance, gain) or labelled-pixel energies; the *mean* column is the mean of the three seed values, computed by us. $\text{gain}_i = \gamma_i / \sqrt{\text{var}_i + \varepsilon}$. The last row is the product of the mean variance with $M + K$, $K = 64$.

Quantity	seed 0	seed 1	seed 2	mean	M
First head BN, median variance	0.1470	0.1467	0.1587	0.1508	48
	0.0678	0.0589	0.0656	0.0641	192
	0.0346	0.0310	0.0398	0.0351	384
First head BN, median gain	2.609	2.611	2.511	2.577	48
	3.842	4.120	3.905	3.956	192
	5.375	5.682	5.010	5.356	384
Second head BN, median gain	2.892	3.122	3.039	3.018	48
	3.110	3.177	3.195	3.161	192
	2.966	3.060	3.121	3.049	384
Pre-BN energy, labelled pixels	25.64	26.28	23.91	25.28	48
	11.07	10.67	11.19	10.98	192
	5.765	5.543	6.015	5.774	384
Post-BN energy, labelled pixels	132.1	130.8	133.2	132.0	48
	130.4	131.5	132.5	131.5	192
	130.7	128.5	130.8	130.0	384
Mean variance $\times (M + K)$	—	—	—	16.9 / 16.4 / 15.7	48 / 192 / 384

Witness Jacobian-vector products. A unit perturbation of the first head convolution is propagated by forward-mode differentiation through the actual head in the actual normalization mode, so that the train-mode derivative includes the dependence on the batch statistics. The final scalar $e_{\text{final}} = N^{-1} \sum_n d_n^\top (\text{diag}(p_n) - p_n p_n^\top) d_n$ is a lower-bound witness for $\lambda_{\max}(G_{\phi\phi})$ and is verified against the same matvec the eigensolver uses; all 216 rows pass at relative discrepancy below 10^{-3} , with a maximum of 1.269×10^{-5} over the 198 rows with nonzero curvature. Table 18 gives the stage energies for two directions in two modes.

Table 18: Witness JVP stage energies, mean over three seeds, from `results/exp1_8d_REPORT.md` Table 2a. `ggn_top` is the full top GGN eigenvector of the convolution block; `s9cen` is the centered leading direction of the incoming patch Gram. e_{pre} is the perturbation energy before the first head normalization, e_{logit} is measured on labelled pixels.

Mode	Direction	M	e_{pre}	e_{bn1}	e_{bn2}	e_{logit}	e_{final}
train	ggn_top	48	260.6	2875	4453	261.7	60.73
train	ggn_top	192	259.4	5905	7553	505.2	118.5
train	ggn_top	384	256.8	9168	13810	1018	227.6
train	s9cen	48	76.53	1538	1224	29.23	5.761
train	s9cen	192	76.53	2305	1914	50.83	9.862
train	s9cen	384	76.53	3997	3906	124.3	21.95
eval_head	ggn_top	48	261.9	261.9	42.42	9.607	2.366
eval_head	ggn_top	192	262.8	262.8	42.18	10.35	2.487
eval_head	ggn_top	384	262.5	262.5	46.53	10.45	2.372
eval_head	s9cen	48	76.53	76.53	9.103	0.4667	0.1049
eval_head	s9cen	192	76.53	76.53	8.460	0.4613	0.1013
eval_head	s9cen	384	76.53	76.53	9.762	0.4450	0.1011

A spatially constant bias perturbation $\Delta b = e_c$ is annihilated exactly by the immediately following train-mode normalization: in `train` and `eval_bn2` modes, at every width and seed, all stage

energies after the centering step are exactly zero and the quadratic form agrees with the matvec to at most 1.14×10^{-17} in absolute value. In `eval_head` the same perturbation gives e_{final} between 7.8×10^{-4} and 2.0×10^{-3} , and in `eval_bn1` between 0.0154 and 0.0692.

Slopes and attribution. Table 19 gives the log-log slopes against M per seed and pooled, together with the mode-to-mode ratios of levels.

Table 19: Log-log slopes against M over the three widths, per seed and pooled, and mode-to-mode ratios of the levels (each mode’s value divided by its train-mode value at the same width and seed, then averaged over seeds and over the three widths), from `results/expl_8d_REPORT.md`. λ_{WW} is the top curvature of the first head convolution’s weight block; λ_ϕ is the whole spatial block. Three-width slopes are descriptive, not asymptotic exponents.

Mode	slope of λ_{WW}				λ_ϕ	level ratio vs train	
	seed 0	seed 1	seed 2	pooled	pooled slope	λ_{WW}	λ_ϕ
train	0.509	0.785	0.350	+0.613	+0.398	1	1
eval_bn1	0.429	0.634	0.542	+0.563	+0.281	0.903	1.03
eval_bn2	0.544	0.632	0.508	+0.571	+0.253	0.156	0.227
eval_head	-0.112	0.181	-0.084	+0.006	-0.331	0.026	0.037

Three readings, and one non-reading. The positive width trend in the spatial block’s curvature at initialization is sensitive to head batch normalization: holding both head normalizations at their initial running statistics removes it ($+0.613 \rightarrow +0.006$ for the convolution block, $+0.398 \rightarrow -0.331$ for the whole spatial block) while leaving the incoming features unchanged. Either normalization alone preserves the trend ($+0.563$ and $+0.571$), which is evidence against attributing it uniquely to the first one, even though most of the drop in the *level* attaches to the second. The mechanism this supports is a finite-range normalization gain acting on width-diluted activations, over the range of widths measured, and not the width-linear feature-Gram mechanism of the shallow theory. The non-reading: the `eval_head` intervention changes the normalization derivative, the forward logits, the downstream gates and the cross-entropy metric at the same time, so a difference between modes establishes sensitivity to the intervention, not a unique cause; and the observation that the variance never approaches the stabilizer is a heuristic about the regime, not a guarantee.

Parameterization control. Table 20 scales the weights and bias of the first head convolution by c and, in the exact invariance, the following stabilizer by c^2 . The logits are unchanged and the block’s top curvature scales by exactly c^{-2} : the raw Euclidean curvature of a layer feeding a normalization is a parameterization quantity, not an architectural one (van Laarhoven, 2017).

Table 20: Function-preserving scale control at seed 0, from `results/expl_8d_REPORT.md` Table 3. *ratio* is $c^2 \lambda_{WW}(c) / \lambda_{WW}(1)$, which is 1 under exact invariance. *logit dev.* is $\max |\text{logits}(c) - \text{logits}(1)|$, reported both with the stabilizer co-scaled by c^2 and with it held fixed (the implementation’s actual behaviour); λ_{WW} , *ratio* and $\|W\|_F^2$ are taken from the fixed-stabilizer variant, and the co-scaled variant differs from it by at most one unit in the fourth significant figure.

M	c	λ_{WW}	ratio	logit dev. (co-scaled)	logit dev. (fixed ε)	$\ W\ _F^2$
48	1/4	719.4	1.000	0	$9.40 \cdot 10^{-4}$	2.003
48	1/2	179.8	1.000	0	$1.89 \cdot 10^{-4}$	8.014
48	1	44.96	1.000	0	0	32.06
48	2	11.24	1.000	0	$4.63 \cdot 10^{-5}$	128.2
48	4	2.810	1.000	0	$5.84 \cdot 10^{-5}$	512.9
192	1/4	1935	0.9998	0	$3.00 \cdot 10^{-3}$	2.002
192	4	7.560	1.000	0	$1.89 \cdot 10^{-4}$	512.6
384	1/4	1927	0.9991	0	$4.69 \cdot 10^{-3}$	1.999
384	4	7.535	1.000	0	$2.95 \cdot 10^{-4}$	511.8

Eigensolver accuracy. Every row of the width sweep terminates on the combined criterion (relative change of the Rayleigh quotient below tolerance *and* residual below 10^{-3}), never on the iteration

or wall-clock budget. Over all 36 (mode, width, seed) rows the relative residual $\|Gv - \lambda v\| / \lambda$ is at most 3.81×10^{-4} for the convolution weight block, at most 3.81×10^{-4} for the weight-bias block and at most 9.97×10^{-4} for the whole spatial block, and the Rayleigh quotient of the returned iterate agrees with the reported eigenvalue to the printed precision in every row.

E.1.4 CONTROLLED RETRAINING: ORIGINAL PROTOCOL

The pipeline was retrained on breast fold 0 at reduced depth (six transformer layers) with full instrumentation. Two protocols were run and both are reported: the *original* protocol (39 runs), which is the pre-registered one and whose joint-versus-frozen contrast an audit of the training code later showed to be confounded, and the *matched* protocol (21 runs), which removes the confounds, saves the selected checkpoint and re-runs the pre-registered comparisons. Sixty logged runs in total. Per-step residuals, block gradient norms, per-epoch parameter displacements and input-Jacobian norms are recorded in both; the matched protocol additionally logs the per-step clipping coefficient and the per-block applied update norms.

The original protocol comprises arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\} \times$ three seeds under AdamW for 60 epochs, plus a {joint-linear, frozen-random} $\times$ three-seed pair under momentum SGD at both widths and a frozen-PCA SGD triple at $M = 48$ as a positive control. The spectral parameters sit in a zero-weight-decay group in every arm, and frozen arms log counterfactual spectral gradients. Table 21 gives the AdamW arms.

Table 21: Controlled study, original protocol, AdamW arms: validation macro-F1 (mean $\pm$ sd over three seeds) under two checkpoint policies, and relative spectral displacement at end of training. *Final-5* is the pre-registered metric. *Best-val* is the maximum over the 60 per-epoch evaluations on the same 28 validation cores; no checkpoint was saved at that maximum and this study has no test set, so that column is exploratory and selection-optimistic. The two policies order the arms oppositely. These arms also differ in clip scope and in normalization-affine treatment, so this contrast is confounded.

Arm	Final-5 (pre-registered)		Best-val (exploratory)		$\ \Delta\theta\ / \ \theta_0\ $
	$M=48$	$M=192$	$M=48$	$M=192$	
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.873 ± 0.030	0.858 ± 0.056	0.54–0.64
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0.840 ± 0.025	0.817 ± 0.017	0
Frozen PCA	0.731 ± 0.003	0.726 ± 0.012	0.769 ± 0.021	0.862 ± 0.068	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.900 ± 0.000	0.897 ± 0.001	0.85–0.96

On the pre-registered final-window metric the joint-linear minus frozen-random gaps are -0.110 at $M = 48$ (90% paired- t interval $[-0.333, +0.114]$) and -0.085 at $M = 192$ ($[-0.185, +0.015]$), neither resolvable at $n = 3$; against frozen-PCA they are -0.193 ($[-0.334, -0.052]$) and -0.081 ($[-0.175, +0.014]$). Taking each run’s maximum over its 60 evaluations reverses the ordering: joint minus frozen-random becomes $+0.033$ and $+0.041$, joint minus frozen-PCA $+0.104$ and -0.004 . The switch is an arithmetic identity — the change in every gap equals the difference in peak-to-final degradation between the two arms — and averaged over seeds that degradation is 0.335 ($M = 48$) and 0.212 ($M = 192$) for joint-linear against 0.192 and 0.086 for frozen-random.

Three asymmetries. First, gradient-norm clipping at 1.0 was computed over the optimizer’s parameters, that is over $\theta + \phi$ in the joint arms but over ϕ alone in the frozen arms, so at identical gradients the frozen arm takes the larger ϕ step: the median extra clip factor paid by the joint arm is 1.10–1.12 under SGD at $M = 48$ and 3.0–3.6 under AdamW at $M = 192$. Second, the normalization following the spectral projection contributes 128 affine parameters that are trained in joint-linear, frozen in frozen-random and structurally absent in frozen-PCA. Third, no checkpoint was saved during training, so the best-validation column is a post-hoc curve maximum that no saved model realizes, taken over 60 evaluations of the same 28 validation cores that define the comparison.

E.1.5 MOMENTUM-SGD CONTROLS AND NORMALIZATION RECALIBRATION

The two optimizer families constitute an optimizer-family control; the SGD arm is not an instantiation of the two-timescale theorem’s gradient flow. It uses momentum 0.9; weight decay 0.01 on ϕ

and 0 on θ ; mini-batches of four cores with reshuffling; and global gradient-norm clipping at 1.0, which binds on 73–89% of steps. Logged gradient norms are recorded before clipping, so the plotted theory quantities are not the quantities the trajectory applied. Under momentum SGD the paired joint minus frozen-random gap is +0.002 at $M = 48$ and +0.011 at $M = 192$ on best-validation, with 90% paired- t intervals $[-0.008, +0.011]$ and $[-0.007, +0.030]$; on the final-window metric the same pairs give +0.028 and -0.057 , with intervals $[-0.173, +0.228]$ and $[-0.124, +0.010]$ — gaps that differ in sign between the widths, with intervals an order of magnitude wider than the effect they bracket. Both arms lose 0.12–0.22 macro-F1 between their peak and their final window. Descriptively, the spectral share of the summed squared gradient norms drops from about 0.94 (AdamW joint) to about 0.26 (SGD joint); momentum, clipping, weight decay and adaptivity all differ between the two families, so no single factor is isolated.

Recalibration of normalization statistics. On disposable copies of the final checkpoints, with all weights fixed, the normalization running statistics were re-estimated on training data only (29 batches) and the model re-evaluated. For this intervention and these checkpoints, the recovered fraction of the joint arms’ peak-to-final drop is 0.17 at $M = 48$ and 0.07 at $M = 192$: recalibration recovers a nonzero part of the degradation but not most of it. What it does undo is the instability of the endpoint number: it repairs the two collapsed joint checkpoints (+0.24 and +0.15 macro-F1) and shrinks the seed standard deviation by roughly 3.7 $\times$, while leaving the frozen arms essentially unchanged ($|\Delta| \leq 0.02$, and ≤ 0.006 for frozen-PCA, whose spectral stage carries no normalization).

E.1.6 MATCHED PROTOCOL

The matched protocol equalizes the three asymmetries and re-runs the pre-registered comparisons at $M = 48$: the reduction-stage normalization affine parameters are placed in ϕ and trained wherever that normalization exists; the clip norm at 1.0 is computed over ϕ only under the same clipping rule and scope in every arm; and the best-validation checkpoint is saved. AdamW at learning rate 10^{-4} , 60 epochs, three seeds, flip and 90°-rotation augmentation on in all arms. The logged per-step clip coefficient averages 0.33–0.57 across arms — the coefficients and gradients differ by arm, which is an outcome of the matched rule — and the logged applied update norms confirm that the spectral share of the applied update tracks the learning-rate multiplier (0.02, 0.17 and 1.36 times the ϕ update norm for $\times 0.1$, $\times 1$ and $\times 10$). Table 22 gives all seven arms.

Table 22: Controlled study, matched protocol ($M = 48$, AdamW, three seeds, mean $\pm$ sd). Identical normalization-affine treatment and the same ϕ -only clipping rule and scope in every arm; best-validation checkpoints saved. *Final-5* is the pre-registered metric; *best-val* is the maximum over 60 per-epoch evaluations on the 28 fold-0 validation cores and is exploratory and selection-optimistic, since this study has no test set. The drift column is the relative spectral-parameter displacement at end of training; its standard deviations were computed by us from `results/e3c_analysis_runs.csv`. Rows 1–2 and rows 3–4 are the two matched pairs; see the text.

Arm	Final-5 (pre-registered)	Best-val (exploratory)	$\ \Delta\theta\ / \ \theta_0\ $
Frozen random	0.684 ± 0.037	0.814 ± 0.040	0
Joint (linear)	0.694 ± 0.056	0.837 ± 0.029	0.566 ± 0.026
Frozen pretrained	0.690 ± 0.033	0.861 ± 0.022	0
Fine-tuned pretrained	0.710 ± 0.052	0.857 ± 0.009	0.020 ± 0.005
Joint, spectral lr $\times 0.1$	0.626 ± 0.055	0.841 ± 0.007	0.083 ± 0.006
Joint, spectral lr $\times 10$	0.561 ± 0.114	0.886 ± 0.037	4.910 ± 0.402
Joint, cosine schedule	0.660 ± 0.020	0.825 ± 0.033	0.359 ± 0.013

Two matched pairs, not a single-factor design. The four core arms do not all differ in one factor. Joint-linear and frozen-random use a linear reduction stage with normalization; frozen-pretrained and fine-tuned-pretrained use a pretrained multilayer spectral encoder without reduction-stage normalization. The clean freezing comparisons are therefore the two matched pairs, joint-linear against frozen-random and fine-tuned against frozen-pretrained. Comparing across the pairs — frozen-pretrained against frozen-random, for instance — changes architecture, initialization and training history, and validation selection, not only informativeness.

Verdicts. All comparisons in Table 23 are paired by seed with 90% t -intervals ($t = 2.920$, $n = 3$), conditional on one validation split, and are descriptive seed summaries rather than a population or multi-factor causal analysis.

Table 23: Matched protocol: paired differences with 90% paired- t intervals ($n = 3$). The last two rows compare the two protocols and are reported as descriptive seed summaries; saving a checkpoint changes what is available to evaluate, not the final-window trajectory.

Contrast	Final-5	Best-val
Fine-tuned – frozen-pretrained	+0.021 [−0.022, +0.063]	−0.004 [−0.048, +0.039]
Joint-linear – frozen-random	+0.011 [−0.142, +0.163]	+0.023 [−0.066, +0.112]
Frozen-pretrained – frozen-random	—	+0.047 [−0.042, +0.136]
Spectral lr $\times 10 - \times 1$	−0.133 [−0.419, +0.152]	+0.048 [−0.016, +0.112]
Spectral lr $\times 0.1 - \times 1$	−0.068 [−0.173, +0.037]	+0.004 [−0.039, +0.046]
Joint/frozen gap: matched – original	+0.120 [−0.230, +0.470]	−0.010 [−0.105, +0.085]

The pre-registered prediction that fine-tuning a pretrained spectral encoder falls below leaving it frozen is not supported: neither difference is resolvable at $n = 3$, and the fine-tuned arm moved θ by only 2% of its initial norm. The logged applied update norms verify nonzero applied encoder updates under AdamW in that arm; this is a statement about the applied updates, not an attribution of the loss decrease. The pre-registered harmful-drift ordering $\times 0.1 \geq \times 1 \geq \times 10$ is not observed: on best-validation the $\times 10$ arm is above $\times 1$ in 3/3 seeds while its final-window endpoint is the lowest and most variable of any arm at 0.561. Three multiplier means do not establish a systematic relation between displacement and peak score, and the intervals are wide. The cosine schedule improved neither mean endpoint (0.825 best-val and 0.660 final-5 against 0.837 and 0.694 for the constant learning rate); one schedule that did not help does not rule out an excessive late learning rate.

Disclosures. Width 48 only; $n = 3$; a single validation fold of 28 fold-0 cores; no test set. The best-validation column is exploratory and the final-window column is the pre-specified metric. The pretrained encoders were selected by pixel validation macro-F1 on the *same* fold-0 validation cores the runner later scores, which is a selection advantage for both pretrained arms, and were pretrained with weight decay 0.01 on θ . Pixel pretraining used the unaugmented centre-crop protocol while subsequent training used random crop plus augmentation; this affects comparisons across the two pairs, not the frozen-versus-fine-tuned pair that starts from the same checkpoint. Augmentation is on in all arms. Finally, the summary artifact `results/e3c_analysis_runs.csv` does not expose per-run configuration hashes or data-order pairing, so before the pairs are described as completely matched a compact provenance manifest should accompany it, recording architecture, initial checkpoint hash, parameter-group membership, clipping and weight-decay settings, normalization mode and state, augmentation and data-order seed, scheduler and selection rule; identical seed labels alone do not establish bit-identical mini-batch order across different model constructions.

E.1.7 WHAT IS NOT CERTIFIED ON THIS PIPELINE

The spatial module’s input-Jacobian operator norm grows over training, from 3.2 to 157 on average for joint-linear (up to 414) and from 1.7 to 1.2×10^4 for joint-MLP, which rejects an initialization-scale stability approximation for the containment constant of the displacement theorem, though not the existence of a finite-horizon constant over the training window. The fitted per-step residual-envelope rate for joint-linear grows from $\hat{\mu} = 6.0 \times 10^{-4}$ ($M = 48$) to 1.9×10^{-3} ($M = 192$), a factor 3.2 for a $4\times$ width increase; the ratio is invariant to the step-clock-to-flow-time conversion, the absolute values are not, and two widths do not establish a trend.

F VERIFICATION RECORD

The statements of this paper rest on the following checks, each a script in the accompanying repository with fixed seeds and a recorded outcome; the record is scoped to these statements and does not certify anything else. (i) Theorem 1: the closed forms, bounds and the normalized-readout control were verified against a full $(2 + M)$ -parameter integration of the flow

over 525 non-trivial initialization-width-margin cases and a 4,000-draw Gaussian initialization sweep (code/experiments/toy_serial_ce.py, toy_serial_ce_isotropic.py; Appendix B); the algebraic identities of the proofs were checked symbolically and numerically (code/audits/astra_math_02_checks.py). (ii) Experiment 2: the matrix-product implementation of the convolutional head agrees with nn.Conv2d with circular padding to 10^{-15} in double precision, in outputs and parameter gradients, at widths 2, 37 and 512 (exp2_intervention.py -selftest); the calibration is bitwise reproducible before and after the paired-condition change to the generator; the pre-registered constants, thresholds, predictions and every amendment are dated in paper/REFOCUS_PLAN_2026-09-10.md. (iii) Experiment 3: preprocessing, principal directions and the readiness scan are computed on training patients only and evaluated on a core-disjoint half; no test-patient quantity was used for any choice. (iv) The production-model measurements of Appendix E carry their own verification record from the earlier draft (code/audits/README.md); only the statements quoted in Section 6 are relied upon here.